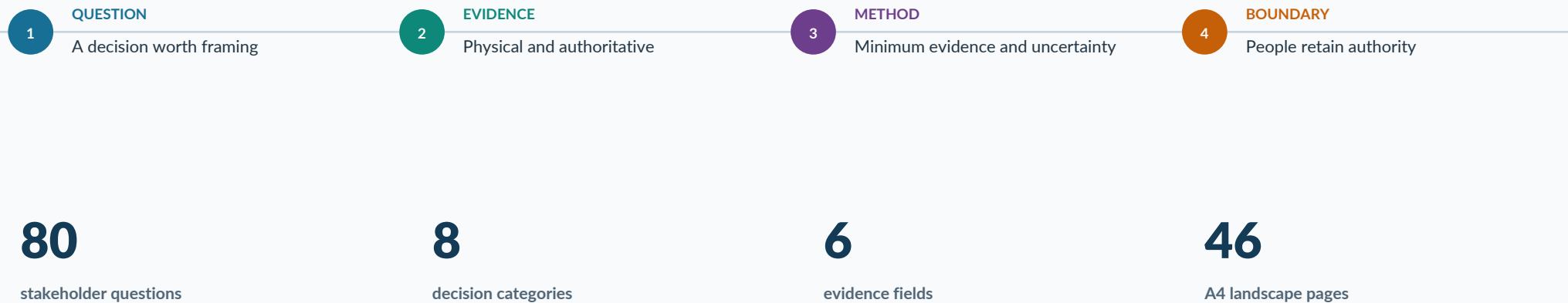


STAKEHOLDER 33

Stakeholder question catalogue

Finance and procurement teams

Eighty practical questions for evidence-led budgeting, purchase planning, contract management, whole-life cost, responsible procurement and bounded financial forecasting in a complex university building.



Evidence support, not delegated authority

The service may organise permitted evidence and calculations. Finance, Procurement, Legal, competent professionals and accountable owners make and authorise decisions.

Use the catalogue to frame answerable decisions, assemble proportionate evidence and route action to accountable owners.

WHAT THIS CATALOGUE SUPPORTS

- 1 **Budget and expenditure context**
Separate price, quantity, scope, timing and accounting effects; expose free balance, allocation and cut-off evidence.
- 2 **Demand and purchasing**
Right-size requirements, test alternatives, expose site dependencies and plan realistic competition, delivery and commissioning.
- 3 **Contracts and invoices**
Compare service levels, rates, milestones, variations, remedies and close-out evidence without making commercial decisions.
- 4 **Whole-life and utilities**
Compare maintenance, replacement, utility, carbon and responsible-procurement evidence across explicit boundaries and time horizons.
- 5 **Forecasts and benefits**
Show scenarios, sensitivity, uncertainty, evidence value and alerts without presenting estimates as promises or spending authority.

EVIDENCE STATUS AND AUTHORITY

- 1 **Proposed by default**
Sensors, integrations and analytics are usable only after deployment, mapping, commissioning, permission and current health are independently verified.
- 2 **Bounded by physics**
A sensor can corroborate only its mapped asset or location, measured phenomenon and covered time. Blind spots and residuals remain visible.
- 3 **Authoritative precedence**
Posted finance records, effective signed terms, current owner registers and competent professional decisions govern when sources disagree.
- 4 **Historical spatial reference**
The May 2021 floor plan is historical only. It cannot establish current room use, access, services, hazards, closures or operating conditions.
- 5 **Safe failure**
Missing, stale, conflicting, unhealthy or unauthorised evidence produces an evidence-gap statement and a routed next action, not invented advice.

The catalogue does not approve budgets, purchase orders, tenders, suppliers, invoices, payments, variations or contracts, and it does not certify value or compliance.

Each card keeps the decision, evidence quality, analysis, authority and practical action visible together.

1 Stakeholder question

A natural finance or procurement decision stated at the level where an accountable person can act.

2 Sensors and telemetry

Proposed modality and placement, asset or location coverage, timing and aggregation, calibration, clock alignment, health, gaps, drift, provenance and physical limits.

3 Authoritative supporting data

Named source owners and precedence, effective, observed and retrieval times, permissions, record versions, staleness and conflict routing.

4 Analysis required

Explicit lookup, observation, calculation, estimate, forecast, comparison, diagnosis or bounded recommendation, with decision interval, minimum evidence, method, uncertainty and safe failure.

5 Answer boundary

What the service must not infer, disclose, promise, certify, approve, authorise or control, and who retains authority.

6 Why this is useful

The concrete budget, sourcing, evidence, sequencing, commercial or escalation action the stakeholder can take.

READING RULE

An answer is decision-ready only when its evidence perimeter, source authority, uncertainty and safe failure are as visible as its calculation.

Eight rules prevent plausible-looking data from becoming unauthorised or overconfident advice.

01 Source precedence

Posted finance records, effective signed terms, current owner records and competent professional decisions govern. Telemetry corroborates and never silently replaces authority.

02 Three decision times

Keep effective time for a rule or term, observed or transaction time for evidence, and retrieval time for the answer. A current dashboard may contain stale evidence.

03 Physical perimeter

State modality, placement, asset or location, cadence, aggregation and blind spots. Historical spatial material cannot establish current operational facts.

04 Quality and lineage

Expose deployment and mapping, commissioning, calibration or validation, clock offset, uptime, gaps, resets, drift, source version and every transformation.

05 Minimum evidence

Every method states identity, completeness, comparison period, owner confirmation and any sample threshold. Below it, the answer becomes an evidence gap.

06 Quantified uncertainty

Show ranges, residuals, confidence intervals, scenario percentiles, coverage and unresolved exposure where supportable. Never replace a range with false precision.

07 Permission and minimisation

Use the least granular permitted data. Do not identify or track people or inspect research content, code, speech or network payloads.

08 Safe failure

When evidence is missing, stale, conflicting, unhealthy or outside scope, suppress the affected conclusion and route the next authoritative action.

FP-001

STAKEHOLDER QUESTION

For each material building cost line, what changed because of unit price, measured quantity, timing, accounting treatment or authorised scope?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor is authoritative for ledger movement. Optional corroboration may use commissioned electricity, heat, gas or water meters and asset run-hour counters linked to the exact charged service. **Coverage, timing and aggregation.** Only mapped meter channels and assets serving the stated cost line and comparison period are in scope; shared or unmetered loads remain a visible residual. Keep native intervals, normally 1-15 minutes, and aggregate to aligned day and accounting month after clock checks. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot distinguish price, coding, accrual or scope effects and must not be used to explain an unmapped cost.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns the general ledger, budget structure, accounting calendar and posting rules; Procurement owns purchase orders and contract prices; Energy and Estates own utility and asset evidence; budget holders own authorised scope changes. Posted finance records and effective signed terms prevail over operational observations; current owner records prevail over the May 2021 floor plan. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat late postings, superseded tariffs, unmatched credits, unapproved scope notes or telemetry with less than the agreed period coverage as stale or incomplete. Show conflicts side by side without silent reconciliation and route them to Finance business partner, Procurement contract owner, Energy manager and relevant budget holder.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and variance decomposition. **Decision interval.** Monthly, with a rolling 12-month and prior-year comparator. **Minimum evidence.** Reconciled actuals for both periods, effective prices, comparable quantities or defensible drivers, approved scope changes, accounting-calendar alignment and at least 90% telemetry completeness where volume data is used. **Method.** Normalise scope and days, then decompose the GBP movement into price, volume, timing, accounting and scope components; leave an explicit residual rather than forcing attribution. **Uncertainty.** Show each component in GBP and percent, the residual, source coverage and a low-high sensitivity range where price and volume interact. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** When comparability or source linkage fails, state which line cannot be decomposed and report only the verified total movement.

ANSWER BOUNDARY

Do not describe a correlation as a cause or infer waste, fault, staff behaviour or budget-holder performance from spend movement alone. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can route each material variance to the correct owner and separate price or scope effects from operational demand.

FP-002

STAKEHOLDER QUESTION

What is the decision-ready free balance after actuals, open commitments, accruals, approved variations, unavoidable forecasts and funding restrictions?

SENSORS AND TELEMETRY

Proposed modality and placement. No physical sensor determines budget availability. Optional delivery scans, commissioning timestamps or work-order status events may corroborate whether a material commitment is likely to crystallise. **Coverage, timing and aggregation.** Corroboration is limited to the named order, asset or project milestone; it does not cover off-system commitments or work not linked to an identifier. Use event timestamps at source and summarise by commitment and accounting period, not by person. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Physical progress cannot establish accounting recognition, cancellation rights or permission to release funds.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns budgets, actuals, accruals, reserves and accounting treatment; Procurement owns requisitions, orders and contract commitments; project and service owners own authorised variations and latest cost-to-complete. The posted ledger, approved budget movements and legally effective commitments prevail; operational status is corroborative only. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat an order without current owner status, an accrual without supporting basis, a forecast older than the reporting cycle or a variation lacking authority as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance business partner, budget holder, Procurement and the accountable project or service owner.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and estimate. **Decision interval.** Weekly for active projects and monthly for cost-centre control. **Minimum evidence.** Current budget, posted actuals, open commitments, supported accruals, approved variations, unavoidable forecast items, cancellation status and known funding restrictions. **Method.** Construct a commitment waterfall from budget to free balance; separate firm, probable, optional and disputed amounts and prevent duplicate counting across order, accrual and forecast records. **Uncertainty.** Report verified free balance plus a low-high range, unresolved GBP, age of uncertain items and the assumptions that move the result. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If commitment status or funding restriction is unresolved, label the balance not decision-ready and do not present it as spendable.

ANSWER BOUNDARY

Do not release encumbrances, move budget, determine accounting treatment or represent an estimated balance as authority to commit. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Budget holders can sequence requests using a bounded free balance and resolve commitments that prevent a reliable decision.

FP-003

STAKEHOLDER QUESTION

Which postings need an owner check for cost centre, project, asset, activity or period coding, and what value could be misstated?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor can establish correct financial coding. Asset-tag scans, work-order asset IDs, delivery-location scans and commissioned meter identifiers may provide limited linkage evidence. **Coverage, timing and aggregation.** Use only identifiers mapped to the invoiced asset, delivery point or completed work; generic building-level observations cannot support line-level recoding. Preserve event time and aggregate only to transaction or invoice line; do not derive individual movement histories.

Verification, privacy and lineage. Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed.

Physical limit. A physical match does not prove the intended funding source, accounting period or tax treatment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns the chart of accounts, project codes, posting dates and corrections; Procurement owns order coding; Estates or IT owns current asset records; requesters and budget holders own business purpose. Finance coding policy and authorised transaction records prevail; current asset ownership may corroborate but cannot override funding or accounting rules. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat orphan asset IDs, closed project codes, copied descriptions, split invoices without allocation basis or owner records not effective for the transaction date as suspect. Show conflicts side by side without silent reconciliation and route them to Finance coding owner, budget holder, Procurement and the relevant asset owner.

ANALYSIS REQUIRED

Operation type. Diagnosis, comparison and recommendation limited to a coding query. **Decision interval.** Monthly before close and at material-project checkpoints. **Minimum evidence.** Complete transaction line, order and invoice linkage, valid code tables for the posting date, asset or service owner, business-purpose evidence and a labelled sample of prior corrections. **Method.** Apply deterministic validation rules first, then similarity or anomaly scoring only to rank cases; test precision on known corrections and present the reason for every flag. **Uncertainty.** Show flagged GBP, confidence band or rule strength, estimated false-positive rate, missing-linkage count and unclassified value. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** Where business purpose or owner evidence is absent, create a clarification case rather than proposing a replacement code.

ANSWER BOUNDARY

Do not change ledger coding, infer the requester's intent from description text alone or expose one cost centre's restricted detail to another. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can prioritise high-value coding checks before close and protect management reports, funding claims and asset records.

FP-004

STAKEHOLDER QUESTION

What proportion of current spend is planned, reactive, statutory, contract-fixed, demand-led or genuinely controllable under the agreed taxonomy?

SENSORS AND TELEMETRY

Proposed modality and placement. Optional work-order priority, alarm event and asset-condition telemetry may corroborate whether an intervention followed an observed incident; no sensor establishes statutory, contractual or discretionary status. **Coverage, timing and aggregation.** Only the named asset, work order and event window are covered; building-wide alarms, blind spots and unlinked calls remain outside the classification evidence. Keep event-level timestamps and summarise by work order and accounting month after clock alignment.

Verification, privacy and lineage. Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed.

Physical limit. An alarm or high reading does not prove necessity, urgency, legal duty or absence of prior planning.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns the reporting taxonomy and spend records; Procurement owns contract commitments; Estates and competent duty holders own maintenance plans, incident classification and statutory obligations; budget holders own controllability judgements. Approved policy, effective contract terms and competent-owner classification prevail over telemetry and free-text descriptions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat uncategorised lines, mixed invoices without allocation, retrospective priority changes or statutory claims without current duty-holder evidence as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance, Procurement, Estates duty holder and budget holder.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and classification comparison. **Decision interval.** Monthly and at budget-setting checkpoints. **Minimum evidence.** An approved mutually exclusive taxonomy, line-level spend, contract linkage, work-order or project evidence, statutory basis where claimed and owner-confirmed treatment for material mixed items. **Method.** Apply rule-based classification, allocate mixed lines only with documented drivers, and keep an unclassified bucket; compare current and prior periods on the same basis. **Uncertainty.** Show GBP and percent by class, unclassified GBP, classification confidence and sensitivity to disputed mixed-line allocations. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** When categories overlap or owner evidence is missing, retain the item as mixed or unclassified rather than forcing a category.

ANSWER BOUNDARY

Do not label spend avoidable, poor value or non-compliant merely because it is reactive or controllable, and do not infer individual performance. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Leaders can set realistic savings and contingency expectations without treating fixed or statutory spend as discretionary.

FP-005

STAKEHOLDER QUESTION

Do repeated low-value purchases represent one aggregable requirement, and is the addressable spend large enough to justify category action?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor is authoritative. Optional stock-issue scans, anonymous service-demand counts and asset fault codes may corroborate repeated physical demand where they are mapped to the purchased item family. **Coverage, timing and aggregation.** Coverage is limited to participating stores, mapped assets and recorded requests; direct purchases, substitutions and unscanned consumption remain gaps. Use event records and aggregate by item family, location and month; suppress person-level detail. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Similar descriptions or nearby demand do not prove that requirements are interchangeable or that aggregation is lawful.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns requisitions, orders, frameworks, category definitions and aggregation rules; Finance owns transactions and thresholds; service owners own technical, accessibility and research requirements. Current procurement rules and owner-validated requirements prevail over text clustering or observed consumption. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat inconsistent supplier names, free-text items without specifications, split accounting lines, outdated catalogue mappings or purchases outside the retrieved data perimeter as incomplete. Show conflicts side by side without silent reconciliation and route them to Procurement category manager, Finance and participating service owners.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and recommendation limited to a sourcing opportunity. **Decision interval.** Quarterly and before relevant framework or tender planning. **Minimum evidence.** At least 12 months of permitted line-level purchases, normalised supplier and item identifiers, current thresholds, owner-confirmed compatibility constraints and coverage of the principal buying channels. **Method.** Cluster by verified need and attributes, not description alone; total quantity, spend and transaction cost; test alternative grouping rules and exclude specialist requirements. **Uncertainty.** Report annualised GBP and volume range, channel coverage, false-cluster sample rate and spend excluded because compatibility is unresolved. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the same-need grouping cannot be owner-validated, present candidate cases only and no aggregate quantity.

ANSWER BOUNDARY

Do not use analytics to evade competition rules, mandate participation, erase specialist needs or disclose one team's sensitive budget to another. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can decide whether category work, a framework, catalogue or coordinated lot is proportionate to the addressable spend.

FP-006

STAKEHOLDER QUESTION

Which recurring charges are approaching notice dates without a confirmed owner, current asset or service link, or documented continuing need?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor proves that a recurring entitlement is unnecessary. Optional commissioned asset heartbeats, licence-server aggregate counts or equipment run-hour signals may corroborate technical activity without identifying users. **Coverage, timing and aggregation.** Only the named asset or service endpoint and agreed observation window are covered; offline, standby, shared and safety-reserve functions remain ambiguous. Use privacy-preserving daily or monthly aggregates and retain the original event availability. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** No heartbeat, low activity or low occupancy must not be treated as non-use or lack of business value.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns recurring payments; Procurement owns renewal terms, notice dates and contract records; IT or Estates owns current asset and service registers; the named service owner confirms continuing need. Signed terms, payment records and accountable-owner confirmation prevail; telemetry may only corroborate activity. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing owner, service-to-asset mismatch, cancellation evidence not recorded, status older than the renewal cycle or telemetry outside the contracted service boundary as unresolved. Show conflicts side by side without silent reconciliation and route them to Procurement contract owner, Finance, IT or Estates owner and accountable service owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to owner confirmation. **Decision interval.** Monthly, with alerts before each contractual notice date. **Minimum evidence.** Payment history, effective contract and notice terms, current owner, asset or entitlement linkage, cancellation status and at least one owner response attempt. **Method.** Match payment-to-contract-to-owner-to-asset; classify confirmed need, pending evidence, duplicate candidate, dormant signal or not assessable; annualise only cancellable exposure. **Uncertainty.** Show annualised GBP range, confidence class, unresolved ownership count and the share supported only by technical activity proxies. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** Where notice terms, ownership or service dependency is incomplete, issue a confirmation task and no cancellation suggestion.

ANSWER BOUNDARY

Do not cancel, renew or vary a service, infer individual licence use or equate inactivity with absence of operational, safety or resilience value. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Contract owners can recover ownership and investigate renewal, duplication or dormancy before contractual notice windows close.

FP-007

STAKEHOLDER QUESTION

Which approved non-personal allocation drivers best distribute shared building costs, and how large is the unmetered or common-area residual?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed commissioned utility submeters, plant-service meters, verified opening hours and anonymous zone counts may support aggregate allocation where direct charging is impossible. **Coverage, timing and aggregation.** Coverage must follow the current meter hierarchy, service areas and space register; common plant, circulation and pass-through loads remain an explicit shared pool. Keep native meter intervals and aggregate to zone-day and accounting month; counts must remain non-identifying. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Occupancy proxies do not establish ownership, benefit, productivity or fairness and must not replace an approved allocation policy.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns allocation policy and postings; Energy owns meter hierarchy and reads; Estates owns current space, service-area and asset records; cost-centre owners confirm use of shared services. Approved allocation policy and reconciled total cost prevail; current owner records override the historical May 2021 floor plan. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unmatched meters, changing room ownership, less than 90% interval completeness, unconfirmed common loads or policy not effective for the period as incomplete. Show conflicts side by side without silent reconciliation and route them to Finance, Energy, Estates and affected cost-centre owners.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and recommendation limited to allocation options. **Decision interval.** Monthly or quarterly, with annual policy confirmation. **Minimum evidence.** Reconciled total cost, current approved policy, verified meter and space mapping, driver coverage for the period and owner agreement on the residual pool. **Method.** Charge directly metered cost first; allocate only the approved shared pool with transparent drivers; preserve unallocated residual and compare at least two defensible alternatives. **Uncertainty.** Show each cost centre's GBP range, percentage directly evidenced, proxy share, residual and sensitivity to alternative drivers. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If policy, mapping or coverage is inadequate, withhold granular allocation and state the evidence needed.

ANSWER BOUNDARY

Do not track people, infer attendance or productivity, treat historical room labels as current, or create an allocation policy through analytics. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can choose an explainable allocation basis, disclose the residual and target metering where it would materially change charges.

FP-008

STAKEHOLDER QUESTION

Which commitments are likely to miss an accounting or funding cut-off, cluster at year-end or lack sufficient accrual evidence?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor determines accounting cut-off. Optional delivery scans, asset-installation events, commissioning timestamps and work-order milestones may corroborate physical progress for a named commitment. **Coverage, timing and aggregation.** Only linked orders, milestones and locations are covered; partial installation, off-site manufacture, unrecorded acceptance and supplier-side events remain gaps. Use source event time and aggregate by commitment and forecast week; retain late-arriving events separately. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Physical progress cannot establish eligibility, transfer of control, accrual treatment or acceptance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns cut-off, accrual rules and funding periods; Procurement owns order and delivery terms; funders own eligibility conditions; project and service owners own authorised milestones and acceptance. Funder terms, Finance policy and signed commercial dates prevail over project estimates or telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat supplier dates not reconfirmed in the current cycle, unsigned acceptance, stale project schedules, unsupported accrual narratives or ambiguous funding end dates as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance, Procurement, fund owner and accountable project or service owner.

ANALYSIS REQUIRED

Operation type. Forecast, comparison and diagnosis. **Decision interval.** Weekly in the final quarter and at each material funding cut-off. **Minimum evidence.** Current order and contract dates, supplier confirmation, authorised milestone plan, acceptance status, funding rules, accounting calendar and material open commitments. **Method.** Build a commitment critical path and cash or expense profile; model delivery and acceptance ranges; identify timing mismatches and evidence gaps for each material item. **Uncertainty.** Report P10, P50 and P90 timing or a bounded scenario range, GBP exposed at each cut-off, coverage and unsupported accrual value. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If eligibility, acceptance or delivery evidence is absent, state the timing as not supportable and route an owner clarification.

ANSWER BOUNDARY

Do not create or approve an accrual, change a funding date, promise supplier delivery or infer acceptance from presence in the building. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance and project owners can gather the right delivery, acceptance or accrual evidence before accounting and funding cut-offs.

FP-009

STAKEHOLDER QUESTION

Is the proposed cost eligible against the current fund, project, cost centre, ownership and benefit-period rules before procurement starts?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor is relevant to funding eligibility. Telemetry may describe operating context only when the funding rule explicitly requires a measured outcome and the competent owner accepts the measurement basis. **Coverage, timing and aggregation.** Any optional measurement is limited to the funded asset, location, baseline and benefit period named in the rule. Use the required native cadence and pre-agreed aggregation; preserve pre-award and post-award periods separately. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Measured use, energy or occupancy cannot establish allowability, additionality, ownership or correct accounting treatment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance and the fund owner hold eligibility rules, budget codes and accounting policy; Procurement holds the proposed order and contract route; the project owner holds the approved purpose, dates and beneficiary scope. Effective fund terms and written Finance interpretation prevail over historical examples, telemetry or requester assumptions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat expired guidance, missing award amendments, unclear benefit periods, costs incurred before authority or purpose descriptions that do not match the current proposal as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance research or funding specialist, fund owner, Procurement and accountable project owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to an eligibility query. **Decision interval.** Before requisition and again before commitment where terms or dates have changed. **Minimum evidence.** Current fund terms, approved project scope, proposed cost and dates, accounting treatment, beneficiary and asset ownership, procurement route and any required approvals. **Method.** Construct a rule-to-evidence matrix; classify each condition as met, not met, owner judgement or missing; identify date and benefit-period mismatches explicitly. **Uncertainty.** Report eligible, ineligible and unresolved GBP separately, with no probability score where the rule requires owner interpretation. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** When a material condition is unclear or missing, state that eligibility is undetermined and route the exact question to the authoritative owner.

ANSWER BOUNDARY

Do not interpret legal or funder terms beyond documented authority, move costs between funds, or present eligibility as permission to buy. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The requester can correct the funding route or obtain an owner decision before sourcing effort and supplier commitment begin.

FP-010

STAKEHOLDER QUESTION

Is the current management cost view decision-ready, and which unresolved defects could move the reported position above materiality?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor establishes the reliability of a management account. Where physical quantities support material costs, only commissioned meter, stock or asset evidence linked to those lines may be tested. **Coverage, timing and aggregation.** Coverage is assessed against the cost lines and periods represented; missing buildings, meters, suppliers or transaction channels must be stated. Preserve source cadence and report completeness by day, month and data domain. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** High telemetry uptime does not correct stale prices, duplicate postings, missing commitments or wrong financial ownership.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns the management view, reconciliations, materiality thresholds and close status; Procurement, Estates, Energy and project owners own contributing records in their domains. The reconciled ledger and current owner-confirmed records prevail over derived dashboards. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unreconciled control totals, late journals, missing commitment feeds, stale contract or asset mapping, unresolved duplicates, permission gaps or transformations without lineage as defects. Show conflicts side by side without silent reconciliation and route them to Finance report owner and each authoritative contributing owner.

ANALYSIS REQUIRED

Operation type. Diagnosis, calculation and recommendation limited to evidence actions. **Decision interval.** At each monthly close and before a material governance decision. **Minimum evidence.** Ledger control totals, source-to-report lineage, completeness by domain, reconciliation status, current mappings, materiality threshold and owner confirmation for material exceptions. **Method.** Score evidence readiness by completeness, reconciliation, timeliness, ownership and lineage; calculate the maximum plausible GBP effect of unresolved defects; do not average away a critical failure. **Uncertainty.** Show coverage percentages, unresolved GBP, directional bias, age of exceptions and a decision-ready, conditional or not-ready status with reasons. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material control total or source lineage fails, suppress the affected interpretation and present the evidence gap rather than a confidence score.

ANSWER BOUNDARY

Do not certify accounts, conceal known defects, infer that a balanced total is complete, or disclose restricted source detail beyond the permitted decision group. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Decision-makers can use, qualify or withhold the headline and direct correction effort to defects with the largest possible effect.

FP-011

STAKEHOLDER QUESTION

What quantity or capacity should be procured from representative demand, service-level and growth evidence rather than the single highest observation?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed anonymous entrance or zone counts, queue sensors, equipment run-time, stock issues and service-request events may corroborate demand without identifying people. **Coverage, timing and aggregation.** Use only mapped service points, rooms, assets or stores relevant to the requirement; declare blind spots, closures, events, remote use and unmet demand not captured by the modality. Keep event or 1-15 minute data and aggregate to hour, day and teaching or operating week after clock alignment. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Counts show observed use, not latent need, user identity, service quality or entitlement to provision.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns pipeline and purchasing route; the requester and service owner own the need and service level; Estates owns current capacity and location records; timetabling or booking owners hold scheduled demand. Approved service requirements and current capacity records prevail; telemetry corroborates only the covered period and location. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat exceptional-event weeks, closure periods, unvalidated counters, duplicate bookings, cancelled sessions or asset capacity not confirmed for the decision date as incomplete. Show conflicts side by side without silent reconciliation and route them to Procurement, service owner, Estates and the relevant scheduling owner.

ANALYSIS REQUIRED

Operation type. Observation, estimate and forecast. **Decision interval.** Per purchasing cycle, using at least one representative operating term or season. **Minimum evidence.** Current capacity, demand observations across representative high and normal periods, service-level target, lead time, growth assumptions and evidence of unmet or displaced demand. **Method.** Create a demand distribution, remove documented exceptional periods only with owner agreement, test capacity at selected service percentiles and compare buy, share, stage and contingency options. **Uncertainty.** Show median and peak-percentile demand, coverage, confidence interval or scenario band, unmet-demand estimate and sensitivity to growth and closure assumptions. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If representative coverage or a service-level target is absent, provide observed ranges only and no recommended quantity.

ANSWER BOUNDARY

Do not identify users, infer attendance or productivity, convert a demand estimate into an approved quantity, or suppress accessibility and safety capacity requirements. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can right-size the requirement, challenge peak-only demand and retain staged capacity where the evidence remains uncertain.

FP-012

STAKEHOLDER QUESTION

Which existing asset, repair, refurbishment, redeployment, sharing, rental or current agreement could meet the need before new buying?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset heartbeats, run-hours, condition indicators, location scans and environmental storage readings may corroborate whether an existing asset is present and technically available. **Coverage, timing and aggregation.** Only assets with current identifiers, verified location and applicable interface or service envelope are covered; off-register items and unmeasured condition remain unknown. Use event or daily aggregates over a representative demand cycle and retain periods of outage, reservation and maintenance. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Low activity does not prove availability, suitability, ownership, licence rights or absence of another team's need.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates, IT or laboratory owners hold current asset, condition, support and location records; Procurement owns frameworks, rentals and contract rights; Finance owns cost and residual value; the requester owns the functional need. Current owner confirmation, safety status, licence rights and effective contract terms prevail over telemetry and the May 2021 floor plan. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing ownership, expired support, unverified condition, incompatible interfaces, restricted funding, unresolved data wiping or future reservations as blockers. Show conflicts side by side without silent reconciliation and route them to Procurement, current asset owner, requester, Finance and the competent technical or safety owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to feasible options. **Decision interval.** Before specification freeze and before commitment. **Minimum evidence.** Validated need, complete candidate inventory, current owner consent, condition and safety status, compatibility, availability window, transfer or rental terms and comparable cost. **Method.** Apply hard feasibility gates first; then compare available options on service fit, timing, transition cost, risk and whole-life cost using the same horizon. **Uncertainty.** Show candidate coverage, option cost ranges, unavailable or unknown inventory count and sensitivity to downtime, transfer and support assumptions. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If ownership, safety, compatibility or availability is not confirmed, mark the option unavailable rather than assuming redeployment.

ANSWER BOUNDARY

Do not seize or reassign assets, infer that inactivity means surplus, override fund restrictions or present an option as approved by the current owner. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can avoid unnecessary acquisition while protecting service continuity, specialist need and future capacity.

FP-013

STAKEHOLDER QUESTION

Which specification clauses are mandatory and evidence-backed, and which preferences should be removed, optionalised or justified?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed environmental, power-quality, network-capacity or equipment-performance measurements may evidence the operating envelope only where the requirement is measurable and location-specific. **Coverage, timing and aggregation.** Use mapped points at the intended installation location and interfaces; short surveys, seasonal gaps and temporary operating states must be declared. Keep native sampling suited to the phenomenon and summarise with distributions, exceedance duration and test conditions rather than averages alone. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot establish legal necessity, user need, accessibility adequacy or vendor neutrality without authoritative interpretation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The service owner owns functional outcomes; competent safety, accessibility, IT and Estates owners hold mandatory standards and interface constraints; Procurement owns competition and specification rules. Current law, policy, approved standards and owner-validated need prevail over historical specifications, vendor claims and measured convenience. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat copied requirements without rationale, superseded standards, unverified site assumptions, named-brand features without equivalence criteria or measurements outside the intended operating envelope as unsupported. Show conflicts side by side without silent reconciliation and route them to Procurement lead and the relevant service, technical, accessibility and safety owners.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to requirement wording. **Decision interval.** During requirements definition and before tender issue. **Minimum evidence.** Validated service outcomes, current mandatory obligations, interface evidence, representative site measurements where needed, acceptance tests and owner for every material requirement. **Method.** Build a requirement-to-evidence matrix; classify mandatory, performance-based, preference or unresolved; test necessity, proportionality, verifiability and supplier-neutral wording. **Uncertainty.** Report the proportion with authoritative support, measurement coverage, unresolved constraints and sensitivity of competition or cost to each contested requirement. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If no competent owner or acceptance test exists, flag the requirement for clarification and do not treat it as mandatory.

ANSWER BOUNDARY

Do not certify technical fitness, remove a safety or accessibility requirement, write vendor-shaped terms or treat sensor averages as the full operating envelope. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can issue a testable specification that protects mandatory outcomes without paying for unsupported preferences.

FP-014

STAKEHOLDER QUESTION

What is the latest safe procurement start date after approvals, competition, lead time, access, installation and commissioning dependencies?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor sets commercial lead time. Optional delivery scans, site-access events, installation milestones and commissioning timestamps may provide observed duration evidence for comparable work. **Coverage, timing and aggregation.** Coverage is limited to linked prior orders of similar scope, route and site conditions; supplier manufacture and approval time outside connected systems remains unobserved. Use source event timestamps and derive stage durations only after timezone and clock alignment. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Past durations do not promise future supplier, approval, access or commissioning performance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns route and tender timetable; Finance and governance owners hold approval dates; suppliers own quoted lead times; project, Estates and service owners hold access, installation and commissioning constraints. Current policy, signed or quoted commercial dates and accountable-owner constraints prevail over historical duration estimates. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat expired quotations, unconfirmed approval slots, outdated access windows, missing dependencies, incomparable historic orders or supplier dates not reconfirmed in the current market as stale. Show conflicts side by side without silent reconciliation and route them to Procurement lead, budget owner, project owner, Estates and relevant supplier contact.

ANALYSIS REQUIRED

Operation type. Forecast, calculation and recommendation limited to a start-by date range. **Decision interval.** At concept approval, before market engagement and whenever a critical dependency changes. **Minimum evidence.** Confirmed required-by date, procurement route, approval calendar, market lead-time evidence, access windows, installation sequence, commissioning and contingency assumptions. **Method.** Build a dependency network with optimistic, most-likely and adverse durations; calculate the critical path and latest safe start under selected confidence levels. **Uncertainty.** Show P50 and P90 completion dates or bounded scenarios, critical dependencies, float and the effect of each unconfirmed duration. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a critical owner date or market lead time is unknown, report no reliable start-by date and identify the missing confirmation.

ANSWER BOUNDARY

Do not promise delivery, reserve access, commit a supplier, bypass governance or convert a forecast date into authority to begin procurement. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can start at the latest defensible date, expose schedule risk and avoid a rushed route that weakens competition.

FP-015

STAKEHOLDER QUESTION

Which requirements can be aggregated without erasing specialist, accessibility, research, location or interface differences?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed stock issues, asset-capacity use and anonymous service-demand counts may corroborate common demand volumes where item and service mappings are stable. **Coverage, timing and aggregation.** Only participating teams, mapped item families and observed demand channels are covered; unrecorded need, specialist exceptions and future projects remain outside. Use event records and aggregate by compatible attribute set and month, never by individual. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Similar volumes do not prove technical interchangeability, common timing or lawful disclosure between teams.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns pipeline, category taxonomy and competition route; each requester owns validated requirements; Finance owns budget and funding restrictions; technical, accessibility and research owners validate non-substitutable needs. Owner-validated mandatory attributes and current procurement rules prevail over clustering or common descriptions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat draft requirements, duplicate pipeline entries, incompatible funding conditions, obsolete specifications or unresolved specialist constraints as barriers to aggregation. Show conflicts side by side without silent reconciliation and route them to Procurement category manager, Finance and participating accountable owners.

ANALYSIS REQUIRED

Operation type. Comparison, calculation and recommendation limited to candidate grouping. **Decision interval.** Quarterly pipeline planning and before market engagement. **Minimum evidence.** In-scope requirements, comparable attribute definitions, required dates, minimum quantities, funding restrictions and owner confirmation of mandatory differences. **Method.** Group only on verified compatible features; quantify common core, optional variants and timing overlap; model lot structures that retain competition and specialist choice. **Uncertainty.** Show demand percentiles, grouped and excluded value, false-grouping sample rate, sensitivity to growth and the effect on supplier participation. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** Where safety, funding, accessibility or compatibility remains unresolved, keep the demand separate.

ANSWER BOUNDARY

Do not mandate participation, disclose one team's restricted budget, erase specialist need or treat aggregation as authority to combine orders. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can aggregate genuine common need while retaining differences that affect compatibility, inclusion or service delivery.

FP-016

STAKEHOLDER QUESTION

Which lot, option, alternative and interface structure preserves competition while controlling integration and lock-in risk?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed protocol and version inventories, interface-status telemetry, power or load characteristics and maintenance-demand records may evidence the installed environment at asset or system level. **Coverage, timing and aggregation.** Only documented interfaces and mapped assets in the intended integration boundary are covered; proprietary internals, future upgrades and undocumented dependencies remain unknown. Use configuration snapshots plus representative operating observations, retaining effective version dates and change history. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Installed prevalence does not by itself justify a proprietary requirement or prove future interoperability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. IT, Estates and competent technical owners hold current architecture, interface and support standards; Procurement owns lotting, alternatives and competition requirements; Legal owns interpretation of licence and dependency terms. Approved outcomes and current interface constraints prevail over vendor preference or inferred installed-base dominance. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat undocumented interfaces, end-of-support versions, contradictory standards, vendor claims without tests, or requirements not linked to an acceptance method as unresolved. Show conflicts side by side without silent reconciliation and route them to Procurement, competent technical owner and Legal.

ANALYSIS REQUIRED

Operation type. Comparison, diagnosis and recommendation limited to commercial structure. **Decision interval.** Before procurement strategy approval and tender issue. **Minimum evidence.** Validated outcomes, current dependency map, interface specifications, transition constraints, market evidence, acceptance tests and rationale for each mandatory compatibility clause. **Method.** Compare lot and option structures against competition, integration, transition and contract-management objectives; identify avoidable bundling and test open-equivalent routes. **Uncertainty.** Show supplier coverage by option, lock-in sensitivity, transition cost range, unresolved interfaces and market-evidence age. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If interface evidence or market capability is inadequate, use outcome-based options or a pilot rather than a closed mandatory architecture.

ANSWER BOUNDARY

Do not favour or exclude a supplier, inspect network payloads, write proprietary requirements without owner authority or certify that a lot structure is legally compliant. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can preserve competition and interoperability while making necessary integration and option costs explicit.

FP-017

STAKEHOLDER QUESTION

Which current site conditions and enabling works must appear as common bidder assumptions and separately priced dependencies?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed competent surveys may use laser dimensions, electrical measurements, structural monitoring, temperature and humidity logging, noise readings, access counts or utility tracing for the specific work area. **Coverage, timing and aggregation.** Coverage must match the installation zone, route, interfaces and enabling-work boundary; the May 2021 floor plan is historical only and cannot establish current conditions. Use sampling appropriate to the condition, including representative occupied and out-of-hours periods where relevant; retain survey date and instrument settings. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Sensing cannot certify structure, asbestos, fire strategy, hidden services, access permission or construction suitability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates owns current building, asset, access and services records; competent surveyors and duty holders own verified site findings; the project owner owns scope; Procurement owns tender assumptions and allocation of risk. Current competent surveys, permits and owner records prevail over historical drawings, generic site descriptions and telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat surveys outside their validity period, inaccessible areas, unresolved intrusive findings, missing utility isolation data or condition changes after the survey as open assumptions. Show conflicts side by side without silent reconciliation and route them to Project owner, Estates, competent survey or safety owner and Procurement.

ANALYSIS REQUIRED

Operation type. Observation, lookup and comparison. **Decision interval.** Before tender issue and after any material site change. **Minimum evidence.** Confirmed work boundary, current surveys, access and outage constraints, service interfaces, known hazards, required permits and a tender assumption schedule. **Method.** Create a condition-to-scope matrix; separate measured fact, authoritative record, bidder verification item, provisional quantity and excluded unknown; price scenarios only where authorised. **Uncertainty.** Show survey coverage, date, inaccessible areas, provisional GBP range and sensitivity to the principal unknown enabling works. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a safety-critical or price-material condition is unverified, identify it as a pre-tender evidence need or explicit bidder assumption.

ANSWER BOUNDARY

Do not certify site safety, hidden conditions, structure or services, grant access, issue permits or transfer risk through an unsupported statement. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Bidders can price from one evidence base, reducing hidden contingency, avoidable exclusions and late variation disputes.

FP-018

STAKEHOLDER QUESTION

What reorder point and lot size protect service continuity at the lowest credible inventory, expiry, storage and cash risk?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed barcode or RFID stock issues, periodic counts, storage temperature or humidity and equipment-consumption counters may corroborate inventory movement and condition. **Coverage, timing and aggregation.** Coverage is limited to mapped stores, item codes and monitored storage zones; informal stock, substitutions, emergency issues and unscanned returns remain gaps. Capture each issue and receipt, reconcile physical counts at an agreed frequency, and aggregate demand by day or week. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Sensor or scan data cannot establish clinical, safety, quality or technical acceptability of stock.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Stores or service owners hold stock records, criticality and shelf-life requirements; Procurement owns lead-time and order terms; Finance owns carrying cost and cash constraints; suppliers own current availability. Current stock count, approved service level and effective supplier terms prevail over inferred consumption. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unreconciled counts, changed item codes, expired shelf-life data, lead times not confirmed in the current market or demand distorted by closures and projects as incomplete. Show conflicts side by side without silent reconciliation and route them to Stores owner, service owner, Procurement and Finance.

ANALYSIS REQUIRED

Operation type. Calculation, forecast and recommendation limited to inventory parameters. **Decision interval.** Monthly for stable items and weekly for critical or volatile items. **Minimum evidence.** Reconciled stock, at least one representative demand cycle, lead-time distribution, minimum order and pack size, criticality, shelf life, storage capacity, carrying cost and substitution rules. **Method.** Estimate demand during lead time, set safety stock against the agreed service level, and compare fixed-order and periodic-order policies under capacity and expiry constraints. **Uncertainty.** Show reorder point and quantity ranges, stockout and expiry probabilities, cash exposure, forecast error and sensitivity to lead-time shocks. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If demand history, stock accuracy or criticality is inadequate, use owner-set minimums and present no optimised quantity.

ANSWER BOUNDARY

Do not place an order, change safety stock for critical items, certify storage conditions or infer individual consumption. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The service owner can protect continuity while reducing emergency buying, expired stock, excess storage and cash tied up.

FP-019

STAKEHOLDER QUESTION

Can the building receive, store, install, commission and support the proposed quantity within the requested delivery window?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed anonymous delivery-bay counts, door or lift state, storage environmental readings, asset-location scans and installation milestone events may corroborate logistics capacity. **Coverage, timing and aggregation.** Use only mapped loading, route, lift, storage and work areas for the planned delivery; temporary closures, shared bookings and manual handling constraints remain explicit. Aggregate access counts to 15-minute or hourly windows and environmental data to the required product tolerance; retain event detail for exceptions. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Observed availability does not grant access, reserve a route, prove load capacity or certify safe handling.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates and Security own current access, routes, lift and loading arrangements; stores and service owners own storage and support capacity; project owner owns installation plan; Procurement owns delivery terms. Current authorised logistics plan, competent capacity checks and booked access prevail over historical floor plans or observed traffic. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unconfirmed route dimensions, unknown floor or lift loads, expired permits, conflicting bookings, environmental readings without calibration or installation labour not committed as blockers. Show conflicts side by side without silent reconciliation and route them to Project owner, Estates, Security, stores or service owner and Procurement.

ANALYSIS REQUIRED

Operation type. Lookup, observation and comparison. **Decision interval.** Before order placement and before each material delivery tranche. **Minimum evidence.** Confirmed quantity and dimensions, delivery window, current route and load data, bookings, storage capacity and tolerances, installation resources, waste removal and support readiness. **Method.** Perform a constraint check by delivery tranche; calculate space-time occupancy and identify the binding route, storage, labour or commissioning constraint. **Uncertainty.** Show capacity margin, unverified constraints, schedule range and consequences of delayed or split delivery. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a safety, access or load constraint lacks competent confirmation, mark the planned tranche infeasible pending owner action.

ANSWER BOUNDARY

Do not book access, operate lifts, direct deliveries, certify load or storage capacity, or assume a route from the historical floor plan. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can set feasible lots and dates before committing to deliveries the building cannot receive or support.

FP-020

STAKEHOLDER QUESTION

Which staged order, pilot, option or volume band best manages uncertain demand without creating premature commitment?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed anonymous use counts, equipment run-time, booking demand, queue measures and service-request volumes may support demand learning during a pilot. **Coverage, timing and aggregation.** Coverage must match the pilot population, location, service hours and intended scale-up boundary; seasonal effects, novelty and unmet demand remain stated limitations. Use pre-agreed intervals and compare baseline, pilot and follow-up periods with aligned clocks and equal aggregation. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Pilot activity does not prove long-term benefit, represent unobserved users or justify automatic scale-up.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The service owner holds demand and acceptance criteria; Procurement owns option and volume-band terms; Finance owns affordability; project and technical owners hold pilot scope, risks and capacity assumptions. Approved pilot criteria and effective commercial options prevail over post-hoc thresholds or telemetry alone. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat a pilot without baseline, success criteria, representative period, exit terms, scale assumptions or owner acceptance as insufficient for commitment. Show conflicts side by side without silent reconciliation and route them to Procurement, Finance, service owner and project owner.

ANALYSIS REQUIRED

Operation type. Forecast, comparison and recommendation limited to commitment structure. **Decision interval.** At business-case formation, tender design and each staged decision point. **Minimum evidence.** Explicit demand scenarios, pilot baseline and acceptance criteria, option prices, minimum commitments, lead times, scale constraints and exit costs. **Method.** Model staged decisions with scenario-weighted cost, service risk and information gained; compare pilot, fixed quantity, call-off, option and volume-band structures. **Uncertainty.** Show demand range, expected and adverse cost, option premium, probability or scenario of underuse and the value of delaying commitment. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If pilot evidence is unrepresentative or criteria changed after observation, report learning only and no scale recommendation.

ANSWER BOUNDARY

Do not define success after results are seen, promise future volume, exercise an option, approve a pilot or identify individual users. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can buy learning and flexibility deliberately instead of paying early for capacity that may not be required.

FP-021

STAKEHOLDER QUESTION

For each service priority, are acknowledgement, attendance, restoration and resolution service levels met on truly comparable incidents?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed alarm events, BMS fault states, equipment status, access-control door events at plant areas and engineer check-in records may corroborate incident and attendance milestones without identifying building users. **Coverage, timing and aggregation.** Coverage is limited to mapped assets, service areas and contract-relevant milestones; phone calls, remote work, manually reported faults and uninstrumented restoration remain gaps. Preserve event timestamps at source, align systems to a common clock and calculate durations by case, not from daily averages. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** A sensor transition does not prove contractual acknowledgement, competent attendance, root cause or final resolution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the contract owner hold effective service levels, exclusions and remedies; the service desk or CMMS owner holds case timestamps and priority; supplier and accountable technical owner hold attendance, restoration and closure evidence. Effective signed terms and authorised case records prevail over telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat reopened cases, priority changes without reason, paused clocks, missing supplier acknowledgement, clock offsets above tolerance or closure entered before owner acceptance as unresolved. Show conflicts side by side without silent reconciliation and route them to Contract owner, Procurement, service-desk owner, supplier manager and accountable technical owner.

ANALYSIS REQUIRED

Operation type. Observation, calculation and comparison. **Decision interval.** Monthly, with immediate attention to critical cases. **Minimum evidence.** Effective SLA version, complete case history, priority and exclusion basis, aligned milestone timestamps, owner acceptance and at least 20 comparable cases where a performance rate is stated. **Method.** Calculate milestone durations under the contractual clock rules; separate excluded, disputed and censored cases; compare attainment by priority and service without pooling unlike cases. **Uncertainty.** Show attainment percentage with confidence interval, median and upper-percentile time, excluded-case count, clock coverage and disputed GBP exposure where relevant. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If terms, clocks or case history are incomplete, present case facts only and no compliance rate.

ANSWER BOUNDARY

Do not determine breach, apply a service credit, infer engineer productivity or identify individual occupants from access or incident data. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The contract owner can focus supplier discussion on comparable cases and distinguish a recurring pattern from isolated extremes.

FP-022

STAKEHOLDER QUESTION

After controlling for criticality, complexity, access and location, where is supplier performance materially different from the comparable baseline?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset fault codes, run-state recovery, plant-room access events and environmental alarms may corroborate case severity and restoration for mapped work orders. **Coverage, timing and aggregation.** Only assets and locations with current contract mapping are covered; differences in age, hidden condition, client delay and unrecorded remote work must remain explicit. Use event-level case timelines and monthly aggregates stratified by comparable case class. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Telemetry cannot measure workmanship, staff effort, user satisfaction or supplier responsibility for every delay.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The contract owner and Procurement hold service scope, supplier assignment and performance definitions; CMMS and Estates hold work history, priority, asset criticality and access constraints; competent owners validate case complexity. Current contract scope and owner-validated case records prevail over raw averages. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat inconsistent priority coding, mixed suppliers, incomplete asset history, changed scope, unresolved access delay or fewer than the stated sample size in a stratum as insufficient. Show conflicts side by side without silent reconciliation and route them to Contract owner, Procurement, CMMS owner and competent asset owner.

ANALYSIS REQUIRED

Operation type. Comparison, diagnosis and estimate. **Decision interval.** Quarterly and before a commercial decision point. **Minimum evidence.** Complete comparable cases, current scope mapping, case-mix variables, aligned milestones, minimum 20 cases per principal stratum or an explicit small-sample treatment. **Method.** Use stratified rates or a transparent risk-adjusted model; compare like-for-like outcomes and test whether conclusions change under alternative case definitions. **Uncertainty.** Show adjusted and unadjusted results, confidence intervals, sample size, coverage, influential cases and residual differences not explained by available evidence. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** Where case mix or sample size is inadequate, avoid ranking and state which comparison is not supportable.

ANSWER BOUNDARY

Do not create a supplier league table from incomparable cases, infer individual worker performance or treat residual association as proven supplier causation. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can challenge performance fairly and identify whether the material constraint sits with supplier delivery, access or service design.

FP-023

STAKEHOLDER QUESTION

Which service incidents meet the exact evidence and notice conditions for a credit, rebate, contractual remedy or owner escalation?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed alarm start and clear events, asset availability states, service-desk timestamps and commissioning measurements may corroborate the duration and scope of a named service event. **Coverage, timing and aggregation.** Coverage is limited to contract-mapped assets, service windows and measured failure modes; partial degradation, user impact and uninstrumented redundancy remain uncertain. Keep raw event time, aligned system clocks and case-level duration; do not infer entitlement from aggregate downtime alone. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot interpret exclusions, materiality, notice requirements, causation or legal entitlement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement, Legal and the contract owner hold effective remedy clauses, notice conditions and caps; CMMS or service owners hold incident facts; Finance holds billed amounts and prior credits; the supplier may provide counter-evidence. Signed terms and authoritative case evidence prevail; telemetry is corroborative. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat an event outside the clause period, missing notice, unresolved exclusion, unmatched asset, disputed restoration time or superseded contract version as unresolved. Show conflicts side by side without silent reconciliation and route them to Contract owner, Procurement, Legal, Finance and accountable service owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and calculation. **Decision interval.** Per material event and before contractual notice deadlines. **Minimum evidence.** Effective remedy clause, complete event timeline, affected service and asset, contractual clock and exclusions, notice status, billed basis and owner acceptance of facts. **Method.** Build a clause-to-evidence matrix and calculate a provisional amount only for conditions that are documented; separate disputed and excluded periods. **Uncertainty.** Show supported duration, clause coverage, provisional GBP range, unresolved conditions and cap or threshold sensitivity. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a condition requiring owner or legal interpretation is missing, label the case potentially relevant and do not state entitlement or amount as due.

ANSWER BOUNDARY

Do not declare breach, issue a notice, apply a credit, settle a claim or disclose restricted dispute material beyond the authorised case team. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can preserve notice deadlines and assemble the precise evidence needed for accountable commercial consideration.

FP-024

STAKEHOLDER QUESTION

Are all compliant bids normalised to the same scope, quantities, exclusions, assumptions, timing and price basis before ranking?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor is authoritative for bid comparability. Approved site measurements or technical test results may support a common quantity or performance assumption when supplied equally to bidders. **Coverage, timing and aggregation.** Any measurement must cover the tendered asset, location and operating envelope and be distributed under the same information conditions. Use the declared test cadence and summary method, retaining raw results and instrument status. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Sensing cannot fill a bidder omission, alter a submitted bid, prove compliance or justify unequal access to information.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns tender documents, bid submissions, clarifications and evaluation controls; evaluators own scored evidence; Finance owns price normalisation rules; Legal advises on permitted treatment; technical owners hold requirements. Issued documents, formally received bids and authorised clarifications prevail over analyst assumptions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat ambiguous units, hidden exclusions, tax or indexation differences, inconsistent quantities, alternative offers not separately identified or clarifications outside the permitted record as unresolved. Show conflicts side by side without silent reconciliation and route them to Procurement lead, Finance evaluator, Legal and authorised technical evaluators.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and comparison. **Decision interval.** During formal evaluation at the approved checkpoints. **Minimum evidence.** Complete bids, issued requirement baseline, permitted clarifications, common quantity and time basis, tax and currency treatment, evaluation rules and conflict declarations. **Method.** Create a traceable bid-adjustment bridge that preserves submitted price, permitted normalisation and evaluated total separately; compare only bids passing the applicable gateway. **Uncertainty.** Show submitted and normalised totals, adjustment ranges, unresolved exclusions and sensitivity to volume, inflation and option uptake. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material assumption cannot be normalised within the authorised rules, flag the bid as not comparable and route it to the evaluation lead.

ANSWER BOUNDARY

Do not change a bid, invent a clarification, reveal one bidder's information to another, select a supplier or certify legal compliance. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Evaluators can compare like with like and see exactly which normalisation assumptions change commercial ranking.

FP-025

STAKEHOLDER QUESTION

Are minimum volumes, call-off bands and price thresholds aligned with observed demand and the approved forecast range?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed anonymous use counts, stock issues, equipment run-time and service-demand events may corroborate physical demand where the contract item or service mapping is current. **Coverage, timing and aggregation.** Coverage is limited to participating locations, assets and channels; off-contract buying, unmet demand, substitutions and future projects remain explicit. Use event data aggregated by contract item, location and month, with seasonal and closure periods retained. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Observed use cannot establish contractual quantity, future need, funding authority or demand outside the measured perimeter.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns contract bands, commitments and call-off records; Finance owns spend and affordability; service owners hold current demand and planned change; suppliers hold current capacity and lead-time statements. Effective contract terms and owner-confirmed demand plans prevail over telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unmapped items, off-contract purchases, incomplete call-off feeds, changed service plans, exceptional periods or supplier capacity not reconfirmed as incomplete. Show conflicts side by side without silent reconciliation and route them to Procurement contract owner, Finance, service owners and supplier manager.

ANALYSIS REQUIRED

Operation type. Calculation, forecast and comparison. **Decision interval.** Quarterly and before crossing a volume or commitment threshold. **Minimum evidence.** Effective bands and minimums, complete call-offs and spend, at least one representative demand cycle, known plans, lead-time constraints and off-contract coverage. **Method.** Forecast demand by scenario, map it to contractual tiers, calculate under-commitment and over-band exposure, and compare option or re-tender timing without assuming future approval. **Uncertainty.** Show P10, P50 and P90 demand or bounded scenarios, expected GBP, minimum-commitment exposure, band-crossing dates and coverage. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If demand perimeter or contract mapping is weak, present historical call-off only and no forecast alignment conclusion.

ANSWER BOUNDARY

Do not vary volume, exercise an option, promise call-offs, infer individual use or treat forecast demand as an authorised commitment. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The contract owner can act before thresholds are crossed and shape future volume terms around a defensible demand range.

FP-026

STAKEHOLDER QUESTION

Does the requested work fall within an active warranty, maintenance, defects or supplier-remedy obligation before paid scope is raised?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed fault alarms, asset event logs, commissioning test results and run-state history may corroborate when and how a mapped asset failed. **Coverage, timing and aggregation.** Coverage is limited to the exact serialised asset, installed configuration and measured failure mode; hidden damage, misuse, external causes and workmanship remain unresolved. Keep raw event timestamps and pre-fault, fault and post-fault windows at the native cadence. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot interpret warranty scope, prove cause, establish liability or certify that intervention is safe.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and Legal hold effective warranty, defect and maintenance terms; Estates or IT holds asset identity, commissioning, service history and current custodian; the competent owner confirms fault description; Finance holds prior charges. Signed terms, serialised asset records and competent findings prevail over telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing serial numbers, transferred assets, lapsed notice dates, undocumented prior intervention, unclear cause or contract versions not effective on installation as unresolved. Show conflicts side by side without silent reconciliation and route them to Contract owner, Procurement, Legal, competent asset owner and Finance.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to a coverage query. **Decision interval.** Before authorising chargeable attendance or parts, subject to safety needs. **Minimum evidence.** Effective terms, asset identity and installation date, fault evidence, maintenance history, notice requirements, exclusions and proposed paid scope. **Method.** Map each fault and proposed activity to relevant obligations and exclusions; separate urgent safety stabilisation from later commercial responsibility. **Uncertainty.** Show covered, excluded and unresolved scope, notice deadline, potential GBP exposure and confidence in asset-term linkage. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If identity, terms or competent fault evidence is missing, issue an urgent owner check while preserving necessary safety action.

ANSWER BOUNDARY

Do not delay emergency safety action, determine liability, direct the supplier, deny payment or represent a possible obligation as accepted coverage. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The requester can preserve warranty or defects rights and avoid commissioning paid scope already covered elsewhere.

FP-027

STAKEHOLDER QUESTION

Do billed rates, indexation, discounts, rebates and cumulative bands match the terms effective on each service date?

SENSORS AND TELEMETRY

Proposed modality and placement. No physical sensor is relevant to contractual rate validation. Metered quantities may be used only when the invoice price depends on a verified measured unit and the meter is contractually accepted. **Coverage, timing and aggregation.** Quantity coverage must match the billing meter, tariff register and billed period exactly; estimated reads and meter exchanges must be explicit. Use native readings and contractual aggregation with aligned billing boundaries. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Meter data cannot establish the correct rate, index, discount, tax or rebate term.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and Legal hold signed schedules, amendments, indexation clauses and rebates; Finance holds invoices, payments and tax treatment; contract owners hold accepted quantities; recognised index publishers own source values. Effective signed terms and authoritative index values prevail over supplier calculations or dashboard rates. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unsigned amendments, wrong base date, missing index publication, retrospective rate changes, cumulative band data with incomplete call-offs or quantities not reconciled to the billed basis as unresolved. Show conflicts side by side without silent reconciliation and route them to Procurement contract owner, Finance, Legal and the accountable quantity owner.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and comparison. **Decision interval.** Per invoice and at each indexation or rebate checkpoint. **Minimum evidence.** Effective pricing schedule, billed units, accepted quantity, index source and base, cumulative volume, discounts, prior credits, tax basis and amendment history. **Method.** Recalculate each rate and cumulative tier from first principles; bridge contracted, calculated and billed amounts and keep disputed assumptions separate. **Uncertainty.** Show line-level variance, total GBP difference, rounding tolerance, unresolved quantity, index sensitivity and cumulative-band coverage. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If an effective term or accepted quantity is missing, flag the line for clarification and do not state an amount payable or recoverable.

ANSWER BOUNDARY

Do not interpret disputed clauses, alter an invoice, approve payment, apply a rebate or expose one supplier's restricted rates outside the authorised team. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can raise an exact rate query, recover supported differences and avoid both overpayment and unsupported withholding.

FP-028

STAKEHOLDER QUESTION

Which documented supplier-side, client-side or shared dependencies explain delay, and what part remains unproven?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed delivery scans, site-access events, permit timestamps, asset-isolation states, storage conditions and installation milestones may corroborate selected dependencies for a named activity. **Coverage, timing and aggregation.** Coverage is limited to mapped access points, work areas, permits and equipment; off-site manufacture, subcontractor actions and informal instructions remain outside. Preserve event timestamps, clock offsets and dependency sequence; aggregate only after case reconstruction. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Temporal order or co-occurrence does not prove causation, responsibility, excusable delay or contractual entitlement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project or contract owner holds the accepted programme and dependency log; Procurement and Legal hold delay and notice terms; Estates and duty holders own access, permits and isolations; suppliers own their notices and evidence. Effective contract terms and authorised contemporaneous records prevail over retrospective narrative and telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing baseline programme, late-entered milestones, unresolved clock conflicts, informal scope changes, incomplete permit records or supplier evidence not linked to the activity as uncertain. Show conflicts side by side without silent reconciliation and route them to Project owner, Procurement, Legal, Estates and supplier manager.

ANALYSIS REQUIRED

Operation type. Diagnosis, observation and comparison. **Decision interval.** Weekly for critical work and at each delay notice. **Minimum evidence.** Accepted baseline and updates, activity dependencies, contemporaneous notices, access and permit records, supplier evidence, client instructions and aligned event times. **Method.** Construct a time-ordered dependency graph; test necessary preconditions and alternative explanations; label supported, contradicted, shared and unknown causes without allocating liability. **Uncertainty.** Show days associated with each supported dependency, overlap, unresolved float, evidence coverage and a range rather than a single attributed delay. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the baseline, notice or event sequence is incomplete, report facts and unknowns only and avoid cause attribution.

ANSWER BOUNDARY

Do not determine liability, grant time or money, direct a claim, infer worker performance or expose restricted dispute material. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can separate documented dependencies from allegation and focus negotiation on the delay drivers that can be evidenced.

FP-029

STAKEHOLDER QUESTION

Before notice dates close, which operational, commercial, market and dependency risks require an owner decision on renew, extend or compete?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed service availability, fault frequency, capacity, environmental condition and anonymous demand telemetry may corroborate the current operating context for contracted services. **Coverage, timing and aggregation.** Coverage is limited to contract-mapped assets, locations and periods; uninstrumented services, user experience, supplier finances and market capacity are not sensed. Use monthly aggregates with event drill-down, retaining outages, closures and changes in scope. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry does not establish supplier solvency, market competitiveness, legal risk, value or future service need.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns contract dates, options, performance and market route; Finance owns spend and affordability; Legal owns interpretation; service and technical owners hold need, dependency and transition constraints; market evidence comes from permitted current sources. Effective terms and accountable-owner need prevail over historic spend or technical activity. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat stale market intelligence, unresolved performance disputes, undocumented dependencies, end-of-support dates without owner confirmation or demand forecasts outside the service perimeter as uncertain. Show conflicts side by side without silent reconciliation and route them to Procurement lead, Finance, Legal, service owner and competent technical owner.

ANALYSIS REQUIRED

Operation type. Comparison, forecast and recommendation limited to decision options. **Decision interval.** Before notice deadlines and at the approved commercial planning point. **Minimum evidence.** Current need, effective options and notice dates, performance evidence, spend, demand scenarios, dependency map, transition plan, market capacity and material risks. **Method.** Compare renew, extend, compete, insource and transition scenarios against service continuity, competition, cost, lock-in and implementation risk; include a no-decision consequence. **Uncertainty.** Show whole-life cost ranges, risk severity and exposure, market-evidence age, scenario sensitivity and unmitigated dependencies. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If notice terms, current need or feasible alternatives are unconfirmed, present the deadline and evidence actions without a preferred route.

ANSWER BOUNDARY

Do not choose the route, exercise an option, negotiate, select a supplier, certify value or imply that poor telemetry alone justifies termination. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The accountable team can protect notice dates and compare renewal, extension and competition before the available route narrows.

FP-030

STAKEHOLDER QUESTION

Which delivery model - internal, supplier or mixed - has the strongest evidenced fit on full cost, capacity, capability and continuity?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed work-order volumes, asset fault demand, service availability, queue measures and aggregate staff or supplier attendance windows may corroborate workload and coverage without measuring individual productivity. **Coverage, timing and aggregation.** Coverage is limited to the defined service, assets, locations and operating hours; informal work, management effort, training and surge capacity remain explicit. Use case-level events and monthly demand distributions, suppressing person-level identifiers. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Telemetry cannot establish competence, morale, employment implications, supplier capability or quality of judgement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The service owner holds outcomes, current capacity and continuity requirements; Finance holds full cost; Procurement holds supplier options and contract data; HR and Legal hold workforce constraints; competent owners hold capability and safety requirements. Approved service requirements and authoritative cost and workforce records prevail over work-volume proxies. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat incomplete overhead, omitted transition cost, incomparable service scopes, unverified internal competence, supplier assumptions without market evidence or demand distorted by temporary projects as incomplete. Show conflicts side by side without silent reconciliation and route them to Accountable service owner, Finance, Procurement, HR, Legal and competent technical owner.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and recommendation limited to delivery-model options. **Decision interval.** At business-case formation and before contract expiry or material service change. **Minimum evidence.** Common outcome and service level, representative demand, internal capacity and competence, supplier evidence, full direct and indirect cost, transition, continuity and workforce constraints. **Method.** Build like-for-like scenarios for internal, external and mixed delivery; capacity-test peak and failure demand; include mobilisation, management, tools, risk transfer and residual owner duties. **Uncertainty.** Show cost ranges, capacity margin, service-risk scenarios, excluded costs, evidence coverage and sensitivity to demand, pay and supplier pricing. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If outcomes or full-cost perimeter differs between options, state that comparison is not decision-ready and identify the missing basis.

ANSWER BOUNDARY

Do not infer individual productivity, make employment decisions, select a supplier, commit internal resources or certify that any delivery model is compliant or best value. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Leaders can compare complete delivery models rather than an incomplete internal cost against a supplier headline price.

FP-031

STAKEHOLDER QUESTION

Does every material invoice line have an authorised order, verified delivery or completion, and valid acceptance for the billed period?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed delivery scans, serial-number or asset-tag capture, work-order status, commissioning test readings and location-specific service telemetry may corroborate receipt or completion.

Coverage, timing and aggregation. Coverage is limited to the exact order line, asset, delivery point or work area; off-site work, bundled services, manual acceptance and hidden defects remain outside. Use event timestamps and line-level identifiers; aggregate only after the three sources are linked.

Verification, privacy and lineage. Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed.

Physical limit. Presence, a status flag or sensor response does not prove contractual acceptance, quantity, quality or price.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns requisitions, orders, receipts and contract terms; Finance owns invoices and payment status; the authorised receiver or technical owner owns acceptance; project or Estates systems hold work evidence. The authorised order, effective terms and recorded acceptance prevail over telemetry or supplier narrative. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing line identifiers, retrospective receipt, acceptance by an unauthorised role, invoice before effective order, split or bundled lines without allocation, and conflicting quantities as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance accounts payable, Procurement, authorised receiver and accountable technical or project owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and calculation. **Decision interval.** Per invoice before payment action. **Minimum evidence.** Valid order line, effective price and quantity, invoice line, receipt or completion evidence, authorised acceptance, tax basis and credit history. **Method.** Perform line-level order-receipt-invoice matching; preserve partial, over, under and disputed quantities; calculate variance without altering any source record. **Uncertainty.** Show matched and unmatched GBP, quantity and price variance, timing difference, evidence coverage and reason-coded exceptions. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material line lacks authorised acceptance or effective order linkage, hold the analytic conclusion at query required and do not state payable status.

ANSWER BOUNDARY

Do not create acceptance, alter an order or invoice, approve payment, reject a supplier claim or infer quality from presence alone. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Accounts payable can route exact exceptions to the correct owner and avoid duplicate payment or unsupported payment delay.

FP-032

STAKEHOLDER QUESTION

Are claimed labour, materials, plant and equipment quantities consistent with authorised scope and contemporaneous work evidence?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed material issue scans, equipment run-hours, plant-room access windows, work-order events and before-after asset readings may corroborate quantities at aggregate job level. **Coverage, timing and aggregation.** Coverage is limited to the named work order and authorised work area; subcontractor time, preparation, travel, off-site fabrication and unscanned consumables remain gaps. Use event-level timestamps, then aggregate by job, day and item class; never create individual worker traces. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed.

Physical limit. Observed attendance or run-time does not equal chargeable labour, productive time, material consumption or contractual entitlement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the contract owner hold rates, measurement rules and authorised scope; the supplier holds submitted records; the work-order owner, quantity surveyor or competent technical owner holds contemporaneous site and completion evidence; Finance holds billed lines. Effective measurement terms and authorised site records prevail over telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat generic timesheets, missing item codes, equipment without job linkage, retrospective entries, scope changes not authorised or quantities outside the measured work perimeter as unresolved. Show conflicts side by side without silent reconciliation and route them to Contract owner, Finance, Procurement and authorised technical or commercial certifier.

ANALYSIS REQUIRED

Operation type. Comparison, calculation and diagnosis. **Decision interval.** Per material invoice or valuation. **Minimum evidence.** Authorised scope and quantities, effective measurement rules and rates, contemporaneous job evidence, supplier substantiation, accepted deliverables and defined tolerances. **Method.** Compare each billed resource with contractual measure and independent corroboration; use engineering ranges or material balances only where assumptions are documented; sample-check low-value lines. **Uncertainty.** Show billed versus supported range, GBP variance, evidence coverage, unexplained quantity and sensitivity to productivity or wastage assumptions without assigning blame. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** Where the contractual measure or work evidence is absent, state not assessable and request the specific substantiation.

ANSWER BOUNDARY

Do not surveil workers, infer individual productivity, determine fraud, certify quantities or direct deduction from an invoice. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The commercial team can ask proportionate quantity questions tied to authorised scope and contemporaneous job evidence.

FP-033

STAKEHOLDER QUESTION

For every material variation, were scope, reason, authority, price basis and timing recorded before the extra cost was incurred?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor establishes commercial authority. Optional work-area events, access records, asset changes and commissioning timestamps may corroborate when changed physical scope began. **Coverage, timing and aggregation.** Coverage is limited to mapped work and assets linked to the variation; design work, off-site fabrication, verbal instruction and hidden condition remain unobserved. Preserve event timestamps and compare them with instruction, quotation and authority times after clock alignment. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Physical change does not prove instruction, necessity, price agreement or entitlement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project or contract owner holds authorised scope and instruction history; Procurement and Legal hold variation terms and delegated authority; Finance holds budget and commitment; supplier and competent technical owner hold quotation and work evidence. Effective contract terms and authorised written instructions prevail over telemetry and retrospective notes. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat instructions without authority, quotations after work start, missing baseline scope, unpriced provisional work, conflicting timestamps or approval outside delegated limits as unresolved. Show conflicts side by side without silent reconciliation and route them to Project owner, Procurement, Finance, Legal and authorised technical or commercial owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and calculation. **Decision interval.** Per variation before commitment and again before valuation. **Minimum evidence.** Baseline scope, change description and reason, authorised instruction, delegated authority, quotation and pricing basis, budget source, start date and completion evidence. **Method.** Construct a chronological scope-price-authority chain; compare changed quantity and rates with contract rules; separate emergency safety action, instructed work and disputed scope. **Uncertainty.** Show authorised, unauthorised and unresolved GBP, timing gap, contingency effect and assumptions behind any estimate. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If baseline, authority or chronology is incomplete, state that the variation is not decision-ready and do not infer retrospective approval.

ANSWER BOUNDARY

Do not create or backdate authority, approve a variation, determine entitlement, certify price reasonableness or instruct the supplier. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The accountable owner can resolve authority, funding and price basis before additional cost becomes embedded or disputed.

FP-034

STAKEHOLDER QUESTION

Which invoice, order or work-order pairs are credible duplicates of the same underlying good, service or event rather than legitimate staged charges?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor identifies a financial duplicate. Asset fault events, delivery scans and work-order identifiers may corroborate whether two records point to one physical event or genuinely separate activity. **Coverage, timing and aggregation.** Coverage is limited to matched assets, dates, locations and service events; recurring services, split deliveries and phased work can legitimately look similar. Use event-level linkage and accounting-period summaries without person identifiers. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Similarity or one shared alarm does not prove duplicate billing or intent.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns invoices, payments and credits; Procurement owns orders and contract identifiers; CMMS, project or service owners hold work and receipt records; suppliers may hold original references. Authoritative document identifiers, effective orders and accepted work records prevail over similarity scores. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat OCR errors, supplier-name variants, credit and rebill sequences, split milestones, recurring charges and records outside the connected perimeter as ambiguity. Show conflicts side by side without silent reconciliation and route them to Finance accounts payable, Procurement, contract owner and relevant work or service owner.

ANALYSIS REQUIRED

Operation type. Diagnosis, comparison and recommendation limited to a duplicate query. **Decision interval.** Daily for new invoices and monthly across the ledger. **Minimum evidence.** Permitted invoice and order lines, payment and credit status, supplier and bank reference, work or receipt linkage, tax amount and known legitimate recurring patterns. **Method.** Apply exact keys first, then explainable fuzzy matching on amount, date, supplier, reference and event; maintain a labelled validation sample and separate credit-rebill chains. **Uncertainty.** Show candidate GBP, confidence or rule strength, precision on the validation sample, unmatched evidence and duplicate coverage across channels. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If physical or commercial linkage is ambiguous, flag possible duplicate and do not block or recover payment automatically.

ANSWER BOUNDARY

Do not accuse a supplier or employee, freeze payment, merge records, expose bank details or label a case fraudulent. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can investigate the highest-value credible duplicate candidates while protecting legitimate recurring or staged charges.

FP-035

STAKEHOLDER QUESTION

Which committed work orders are stalled, how much value is tied up, and which verified dependency needs owner action?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed work-order events, asset status, delivery scans, plant-area access events, permit timestamps and isolation states may corroborate progress and selected dependencies. **Coverage, timing and aggregation.** Coverage is limited to mapped work orders, assets and controlled areas; off-site supply, design decisions, subcontractors and informal communication remain outside. Use event timestamps and weekly stage summaries, retaining the last verified activity and clock quality. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** No activity signal does not prove supplier delay, inactivity or responsibility.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns commitment value; Procurement and the contract owner hold supplier and order status; CMMS or project owner holds work stage and dependency log; Estates and duty holders own access, permits and isolations. Authorised work records and current owner confirmation prevail over inferred inactivity. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat status not updated in the agreed cycle, milestones without evidence, unmatched order and work IDs, unresolved owner actions or supplier dates not reconfirmed as stale. Show conflicts side by side without silent reconciliation and route them to Work owner, Procurement, Finance, Estates duty holder and supplier manager.

ANALYSIS REQUIRED

Operation type. Lookup, diagnosis and forecast. **Decision interval.** Weekly for material open commitments. **Minimum evidence.** Open commitment, current work stage, last verified event, required dependencies, owner and due date for each dependency, supplier status and cancellation or rephasing terms. **Method.** Calculate age in stage, compare with a relevant duration distribution, and build a dependency chain; classify active, waiting on client, waiting on supplier, blocked, complete-not-closed or unknown. **Uncertainty.** Show committed GBP, days in stage, percentile against comparable work, dependency confidence and amount with no recent authoritative status. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the dependency owner or last verified event is missing, classify unknown and issue a status request rather than attributing delay.

ANSWER BOUNDARY

Do not cancel or expedite work, assign fault, change priority, release commitment or infer worker activity from access events. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance and service owners can release, rephase or escalate stalled commitments using the verified dependency and next owner action.

FP-036

STAKEHOLDER QUESTION

Does each milestone or partial-completion claim map to contracted deliverables achieved and accepted at the valuation date?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset commissioning results, equipment states, material installation scans, environmental tests and location-specific completion events may corroborate measurable deliverables. **Coverage, timing and aggregation.** Coverage is limited to the milestone items and acceptance locations; concealed work, documentation, training, integration and quality defects may not be sensed. Use native test records and event timestamps, preserving pass, fail, unavailable and retest states. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Installed equipment, a live signal or percentage complete in a system does not equal contractual completion or acceptance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the contract owner hold milestone definitions and payment terms; project manager, quantity surveyor or authorised technical owner holds measured progress and acceptance; Finance holds prior valuations and payments; supplier holds claim evidence. Effective milestone terms and authorised acceptance prevail over telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat vague percentage claims, missing acceptance criteria, incomplete documentation, changed scope, tests outside tolerance or duplicate value from earlier milestones as unresolved. Show conflicts side by side without silent reconciliation and route them to Authorised project or commercial owner, Finance and Procurement.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and calculation. **Decision interval.** At each valuation or milestone claim. **Minimum evidence.** Effective deliverable definition, authorised scope, measurable quantities, acceptance criteria, current evidence, prior certified value, retentions and unresolved defects. **Method.** Map claim lines to deliverables and acceptance evidence; calculate cumulative earned value without double counting; keep incomplete and disputed components separate. **Uncertainty.** Show supported value range, claimed and previously recognised GBP, unaccepted deliverables, test coverage and sensitivity to quantity or milestone interpretation. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the milestone is not objectively defined or acceptance is missing, state that the claim cannot be evidenced and do not calculate payable value.

ANSWER BOUNDARY

Do not certify completion, approve payment, accept a deliverable, waive a defect or change the milestone definition. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The authorised certifier receives a traceable claim-to-deliverable map and can identify exactly what remains unaccepted.

FP-037

STAKEHOLDER QUESTION

Did the intervention restore the required service and keep it stable for the agreed evidence window?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset state, environmental condition, alarm recurrence, service availability and performance telemetry may corroborate restoration for the affected service. **Coverage, timing and aggregation.** Coverage is limited to mapped assets, service zones and operating states; redundancy, intermittent demand, user experience and hidden degradation remain explicit. Use native sampling through a defined pre-event, intervention and post-intervention window, with clock alignment and no averaging across outages. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** A normal reading does not prove root cause, workmanship, acceptance, long-term reliability or absence of impact outside coverage.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The service owner and competent technical owner define required service and acceptance window; CMMS holds intervention and closure records; Procurement holds contractual obligations; users may provide authorised service-impact evidence. Owner-defined acceptance criteria and authoritative test results prevail over dashboard appearance. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing baseline, changed setpoints, low demand during the window, unhealthy sensors, unrecorded overrides or recurrence after retrieval as limitations. Show conflicts side by side without silent reconciliation and route them to Competent service owner, CMMS owner and contract owner.

ANALYSIS REQUIRED

Operation type. Observation, comparison and diagnosis. **Decision interval.** After each material intervention and at the end of the agreed stability window. **Minimum evidence.** Defined service metric and tolerance, intervention time, representative demand, healthy mapped telemetry, baseline or expected range, alarm history and owner acceptance. **Method.** Compare pre and post distributions, exceedance duration and recurrence; use change-point or control-chart methods only with sufficient stable data and preserve confounding changes. **Uncertainty.** Show coverage, service-within-tolerance percentage, recurrence count, confidence interval and unresolved confounders. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If demand, sensor health or acceptance criteria are inadequate, state restoration not verifiable and keep the case open for owner evidence.

ANSWER BOUNDARY

Do not certify workmanship, declare root cause, close a work order, accept the service or infer user satisfaction from telemetry. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The owner can distinguish sustained restoration from a temporary reset before deciding the next acceptance or investigation step.

FP-038

STAKEHOLDER QUESTION

Are call-out, travel, standby and overtime charges consistent with the authorised incident, attendance window and effective charging rules?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed plant-area access events, engineer check-in records, alarm timing and asset-state changes may corroborate attendance windows at job level without continuous worker tracking. **Coverage, timing and aggregation.** Coverage is limited to approved access points and the named incident; remote diagnosis, travel origin, breaks, multiple jobs and off-site preparation remain unknown. Use event timestamps and derive only the minimum job-level window needed; suppress individual identity from analytic output. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Entry duration does not equal chargeable time, work performed, travel distance or overtime entitlement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the contract owner hold rate schedules, call-out rules and authorised instruction; Finance holds billed lines; CMMS holds incident timeline; supplier and authorised receiver hold attendance evidence. Effective charging terms and authorised job records prevail over access telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat shared visits, missing check-out, remote resolution, clock conflict, retrospective instruction, bundled travel or overtime without approval as unresolved. Show conflicts side by side without silent reconciliation and route them to Contract owner, Finance, Procurement and authorised service owner.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and comparison. **Decision interval.** Per material charge line. **Minimum evidence.** Effective rate and minimum-charge rules, authorised incident, attendance and work records, billing window, prior related jobs, travel or standby basis and applicable approvals. **Method.** Reconstruct the allowable charging window under the contract; compare billed units and rate; identify overlaps, minimums and shared-call allocation without inferring individual effort. **Uncertainty.** Show supported quantity range, GBP variance, clock uncertainty, overlap and unresolved basis. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the contract rule or attendance basis is ambiguous, raise a clause-specific query and no recommended deduction.

ANSWER BOUNDARY

Do not monitor individuals, infer productivity, calculate personal pay, approve or reject charges, or disclose identifiable access records beyond authorised need. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can query the exact rate, attendance window or authorisation without making unsupported claims about individual activity.

FP-039

STAKEHOLDER QUESTION

What exact evidence supports an invoice query, what value is affected, and which points remain assumptions or unresolved conflicts?

SENSORS AND TELEMETRY

Proposed modality and placement. Use only the modality directly relevant to the disputed line, such as delivery scan, accepted meter read, asset event, commissioning result or work-order timestamp. **Coverage, timing and aggregation.** Coverage must match the invoice line, billed period, asset and location; any uncovered element is stated separately. Preserve raw timestamps and values, contractual aggregation and the evidence snapshot used for the query. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry outside the billing basis, coverage or tolerance must not be presented as contradiction.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns invoice and payment record; Procurement owns order and contract terms; the authorised receiver or technical owner owns delivery, quantity and acceptance; supplier evidence remains part of the record. Effective terms and authorised source records prevail; no derived result silently replaces them. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing line linkage, unresolved unit conversion, source versions retrieved after the decision without effective dating, permission gaps or contradictory owner evidence as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance, Procurement, authorised receiver, contract owner and supplier manager.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to query wording. **Decision interval.** Per exception before communication or payment action. **Minimum evidence.** Invoice line, order and term basis, accepted evidence, calculation, tolerance, source timestamps and owner confirmation of the factual premise. **Method.** Build an evidence statement separating fact, calculation, estimate and unanswered question; reproduce every numeric difference and cite the source record identifiers internally. **Uncertainty.** Show exact GBP or quantity difference, tolerance, evidence coverage, confidence class and the assumption that would reverse the query. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the discrepancy cannot be reproduced or linked, withdraw the numeric claim and request clarification only.

ANSWER BOUNDARY

Do not allege misconduct, disclose unrelated commercial data, issue supplier communication, approve withholding or present a hypothesis as proven error. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can send a narrow, quantified and defensible invoice query that the supplier and accountable owner can resolve.

FP-040

STAKEHOLDER QUESTION

Before close-out and final payment, are retentions, credits, defects, warranties, asset data, manuals and acceptance actions complete and owned?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed commissioning test outputs, asset-tag creation, defect-state events and service telemetry may corroborate that physical assets are present and selected tests occurred. **Coverage, timing and aggregation.** Coverage is limited to mapped deliverables and test points; manuals, training, legal title, hidden defects and document quality are not sensed. Use event timestamps and deliverable-level status, retaining failed, pending and retest outcomes. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** A live asset or passed point test does not prove full completion, document adequacy, warranty validity or financial closure.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project or contract owner holds close-out requirements and acceptance; Procurement and Legal hold retention, warranty and credit terms; Finance holds payments and provisions; Estates, IT and duty holders own asset records and operating documents. Effective terms and authorised acceptance records prevail over system status. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing owner, unsigned acceptance, defects without due date, credit not posted, warranty start conflict, incomplete asset fields or documents not validated by the receiving owner as open. Show conflicts side by side without silent reconciliation and route them to Project owner, Finance, Procurement, Legal and each receiving asset or duty owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and calculation. **Decision interval.** At each close-out checkpoint and before final financial release. **Minimum evidence.** Contract close-out schedule, payments, retentions, defects, credits, warranty terms, accepted deliverables, asset records, manuals, training and named receiving owners. **Method.** Create a deliverable-obligation-payment matrix; reconcile open financial items to required evidence and owner acceptance; separate administrative absence from material service risk. **Uncertainty.** Show open GBP, due dates, evidence completeness, owner gaps, warranty exposure and items blocking safe closure. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material deliverable or owner acceptance is absent, state not ready for close-out and route the specific action.

ANSWER BOUNDARY

Do not release retention, close the contract, waive defects, accept manuals, start or alter warranties, or certify completion. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The accountable team can preserve leverage and prevent missing credits, warranties, asset information or residual obligations after payment.

FP-041

STAKEHOLDER QUESTION

Which assets should receive survey, maintenance, refurbishment or replacement funding first when criticality, condition, risk and whole-life cost are combined?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed vibration, temperature, current, pressure, flow, run-hour, alarm and environmental measurements may corroborate condition for mapped plant and equipment. **Coverage, timing and aggregation.** Coverage is limited to installed sensors and the failure modes they can observe; hidden structure, uninstrumented components, dormant assets and common services remain explicit gaps. Use native condition sampling with daily or weekly health summaries and retain excursions, sensor changes and operating state. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry does not certify condition, safety, statutory status, remaining life or business criticality.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates or IT owns current asset register, condition, maintenance and criticality; competent engineers and duty holders own safety and technical findings; Finance owns cost and funding; service owners own consequence and continuity needs; Procurement owns market and contract evidence. Competent findings and current owner records prevail over model scores and the historical floor plan. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unmapped assets, stale surveys, incomplete work history, changed service criticality, uncoded dependencies or telemetry that misses relevant failure modes as unresolved. Show conflicts side by side without silent reconciliation and route them to Accountable asset owner, competent engineer, Finance, Procurement and affected service owner.

ANALYSIS REQUIRED

Operation type. Comparison, estimate and recommendation limited to a ranked evidence set. **Decision interval.** Quarterly for portfolio planning and before capital allocation. **Minimum evidence.** Current asset identity, condition and criticality, maintenance history, service consequence, feasible options, comparable whole-life costs, lead times and owner risk tolerances. **Method.** Apply hard safety and statutory gates first; then use transparent multi-criteria scoring with scenario whole-life cost and service consequence; test rank stability under alternative weights. **Uncertainty.** Show cost ranges, evidence confidence, score contribution, rank sensitivity, unmapped assets and items that remain owner judgement. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a critical asset lacks competent condition or consequence evidence, place it in urgent evidence needed rather than assigning a reassuring rank.

ANSWER BOUNDARY

Do not certify condition or safety, set organisational risk appetite, approve capital priority or treat a model ranking as a funding decision. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Leaders can direct surveys and funding to assets where consequence, condition and cost justify the earliest accountable attention.

FP-042

STAKEHOLDER QUESTION

Which assets show sustained deterioration in failure, downtime or corrective cost after changes in use and recording are controlled?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed fault events, run-hours, vibration, temperature, pressure, current and service-state telemetry may corroborate deterioration for serialised assets. **Coverage, timing and aggregation.** Coverage is limited to mapped assets, operating regimes and observable failure modes; maintenance resets, sensor replacements, changed alarms and demand shifts must be retained. Use event-level fault and work history with condition data at native cadence, aggregated by run-hour and month. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Rising alarms or cost does not by itself prove physical deterioration, root cause or future failure.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. CMMS or IT service management owns failure, downtime and work history; Finance owns cost; the asset owner holds operating context and criticality; competent engineers own diagnosis; Procurement holds supplier and parts history. Competent findings and validated work records prevail over statistical trend. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat coding changes, merged assets, missing run-hours, changed service hours, unclosed work, cost inflation not normalised or telemetry drift as confounders. Show conflicts side by side without silent reconciliation and route them to Asset owner, competent engineer, CMMS owner, Finance and Procurement.

ANALYSIS REQUIRED

Operation type. Observation, diagnosis and estimate. **Decision interval.** Monthly, with quarterly asset-health assessment. **Minimum evidence.** At least 12 months or 20 comparable events where available, stable asset identity, exposure measure such as run-hours, consistent failure definitions, normalised cost and documented configuration changes. **Method.** Calculate failure and downtime rates per exposure, inflation-adjust cost, apply control charts or change-point tests, and compare with peer assets only after case matching. **Uncertainty.** Show rate ratios, confidence intervals, change date range, data coverage, effect of coding changes and residual uncertainty. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If exposure, identity or definitions changed materially, report an apparent trend with causes unresolved rather than deterioration.

ANSWER BOUNDARY

Do not declare root cause, predict imminent failure, infer maintenance staff performance or authorise replacement from a trend alone. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Asset and finance teams can distinguish real deterioration from noisy records and target competent diagnosis before exposure grows.

FP-043

STAKEHOLDER QUESTION

What remaining service-life range is supportable for each critical asset, and which additional evidence would narrow it most?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed condition and usage telemetry such as vibration, temperature, electrical signature, pressure, cycles and run-hours may inform degradation for mapped critical assets. **Coverage, timing and aggregation.** Coverage is limited to measured components, known operating states and validated failure modes; latent defects, environmental shocks, obsolescence and shared dependencies remain outside. Use native sampling with exposure-based summaries and retain sensor replacements, maintenance resets and abnormal regimes. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot certify remaining life, safety, parts availability or future operating demand.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The accountable asset owner and competent engineer hold condition, duty and failure-mode evidence; CMMS holds history; manufacturers and suppliers hold support and obsolescence data; Finance owns cost horizon; service owners hold future demand. Competent engineering judgement and current support evidence prevail over model output. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unknown age or configuration, sparse failures, changed duty, unvalidated sensors, missing maintenance history or manufacturer estimates outside applicable conditions as uncertainty. Show conflicts side by side without silent reconciliation and route them to Competent engineer, asset owner, service owner, Procurement and Finance.

ANALYSIS REQUIRED

Operation type. Estimate, forecast and comparison. **Decision interval.** Annually and before a material maintain-or-replace decision. **Minimum evidence.** Verified asset age and configuration, duty cycle, condition, relevant failure history, maintenance effects, support horizon, demand scenario and a defensible degradation or survival model. **Method.** Combine engineering limits with survival or degradation scenarios; avoid fitting complex models to sparse events; identify assumptions through sensitivity analysis. **Uncertainty.** Show a percentile or bounded life range, probability of failure by planning horizon, model validation, evidence coverage and top sensitivity drivers. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** Where data or failure physics are inadequate, provide owner-defined planning scenarios and no numerical life claim.

ANSWER BOUNDARY

Do not guarantee life, defer a safety action, certify continued operation, or present a statistical percentile as an engineering decision. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can time surveys, spares and replacement planning against a defensible life range rather than a false single date.

FP-044

STAKEHOLDER QUESTION

Would changing the planned-maintenance interval reduce expected whole-life cost without increasing service, safety or compliance exposure?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed run-hours, cycles, condition indicators, alarm history and post-maintenance performance may support interval analysis for mapped assets and specified maintenance tasks. **Coverage, timing and aggregation.** Coverage is limited to observable failure modes and actual duty; mandatory checks, hidden degradation and tasks without measurable outcomes remain outside. Use exposure-based event history across multiple intervals and retain task dates, resets, sensor changes and operating conditions. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot waive prescribed intervals, certify safety or prove that maintenance caused an observed change.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Competent asset and duty owners hold maintenance strategy, mandatory intervals and risk controls; CMMS owns task and failure history; Finance owns cost; manufacturers and contracts hold recommendations and obligations. Law, approved maintenance policy, warranty and competent-owner limits prevail over cost modelling. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat tasks with changed scope, incomplete completion records, low failure counts, confounded operating duty or intervals outside manufacturer and duty-owner bounds as insufficient. Show conflicts side by side without silent reconciliation and route them to Competent asset or duty owner, Finance, CMMS owner and Procurement contract owner.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and recommendation limited to testable interval options. **Decision interval.** At annual maintenance planning or after material evidence change. **Minimum evidence.** Current mandatory bounds, task cost and downtime, failure and consequence history by exposure, condition evidence, warranty constraints and enough cycles to compare intervals. **Method.** Model preventive and corrective cost under candidate intervals, including downtime and failure consequence; use controlled pilot or phased monitoring where evidence is weak. **Uncertainty.** Show whole-life cost range, failure-risk range, sample size, coverage, assumptions and break-even interval. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a mandatory bound, competent risk judgement or minimum event history is absent, retain the current authorised interval and state the evidence gap.

ANSWER BOUNDARY

Do not alter maintenance schedules, waive statutory or warranty duties, certify risk, or infer that lower observed failure proves a safer interval. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The owner can test evidence-led interval changes while keeping non-negotiable safety, warranty and service duties intact.

FP-045

STAKEHOLDER QUESTION

Which spares should be held locally, centrally or on demand given failure demand, lead time, criticality, obsolescence, storage and cash?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed stock scans, part issues, asset fault events, run-hours and storage environmental readings may corroborate demand and stock condition. **Coverage, timing and aggregation.** Coverage is limited to mapped stores, compatible part numbers and covered assets; cannibalisation, informal stock, substitutions and supplier inventory remain uncertain. Capture receipt, issue and count events and aggregate demand by part family and month; monitor storage conditions at the required cadence. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Issue history does not prove future failure, compatibility, shelf life, genuine provenance or safe substitution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Stores and asset owners hold stock, compatibility and criticality; Procurement holds supplier lead time, price and obsolescence; Finance holds carrying cost; competent technical owners approve substitutes and minimum resilience stock. Current compatibility and criticality decisions prevail over demand models. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unreconciled counts, superseded part numbers, unsupported substitutes, lead times not reconfirmed, storage outside limits or failure demand distorted by a one-off programme as incomplete. Show conflicts side by side without silent reconciliation and route them to Stores owner, competent asset owner, Procurement and Finance.

ANALYSIS REQUIRED

Operation type. Calculation, forecast and recommendation limited to stocking policy. **Decision interval.** Quarterly and before obsolescence or major maintenance events. **Minimum evidence.** Reconciled stock, compatible installed base, failure demand, lead-time distribution, consequence of stockout, shelf life, storage capacity, carrying cost and supplier alternatives. **Method.** Calculate demand during lead time and expected downtime cost; compare local, central and supplier-held options with criticality and obsolescence constraints. **Uncertainty.** Show recommended stock range, stockout probability, expected expiry or obsolescence cost, cash tied up and sensitivity to common-cause failures. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If compatibility, lead time or consequence is not verified, retain owner-set resilience stock and no optimised reduction.

ANSWER BOUNDARY

Do not change safety stock, approve substitutes, dispose of parts, commit supplier inventory or certify storage compliance. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can protect critical continuity without holding every part everywhere or releasing stock on incomplete demand evidence.

FP-046

STAKEHOLDER QUESTION

Among technically compliant options, which has the lowest comparable whole-life cost across purchase, integration, energy, maintenance, downtime, renewal and disposal?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset load, run-hour, environmental condition and maintenance telemetry may establish the current operating profile used in option costing. **Coverage, timing and aggregation.** Coverage must match the service, duty cycle, location and interfaces of the proposed options; future use, vendor performance and unmeasured downstream effects remain scenarios. Use representative seasonal intervals and aggregate to annual duty distributions with raw lineage retained. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry does not establish option compliance, supplier capability, future energy use, maintenance quality or disposal route.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement holds compliant bids and commercial terms; Finance owns discount rate, inflation and cost policy; service and technical owners hold requirements, duty and integration; Energy owns tariffs and measured profile; competent owners hold end-of-life obligations. Only options that pass authoritative mandatory gates enter cost comparison. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat omitted transition, licences, enabling works, residual owner duties, inconsistent life horizons, unverified energy claims or end-of-life assumptions without owner evidence as incomplete. Show conflicts side by side without silent reconciliation and route them to Finance, Procurement, service owner, competent technical owner and Energy or sustainability owner.

ANALYSIS REQUIRED

Operation type. Calculation, estimate and comparison. **Decision interval.** At business-case and tender evaluation checkpoints. **Minimum evidence.** Common requirements, compliant options, complete cash-flow perimeter, service life, operating profile, maintenance and downtime assumptions, residual value, discount and inflation basis. **Method.** Construct discounted cash flows on a common price and time basis; model energy and failure scenarios; separate transfer payments from real resource cost and show excluded benefits. **Uncertainty.** Show net present cost range, annual equivalent cost, break-even points, source coverage and sensitivity to life, demand, energy, downtime and residual value. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If options have different mandatory outcomes or material costs are missing, state they are not comparable and do not rank them.

ANSWER BOUNDARY

Do not certify compliance or value, select an option, approve a business case or treat the lowest modelled cost as the required decision. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Decision-makers can compare the full cost of compliant choices and see which assumptions, not headline price, determine value.

FP-047

STAKEHOLDER QUESTION

Which deferred-maintenance items are accumulating material financial, continuity or duty-holder exposure and need the earliest owner decision?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed condition indicators, alarms, run-hours, failure events and service availability may corroborate current exposure for mapped deferred tasks. **Coverage, timing and aggregation.** Coverage is limited to observable assets and failure modes; hidden degradation, common systems, statutory duties and consequence outside the measured area remain explicit. Use native condition data and monthly task-age summaries, retaining operating state and sensor health. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Absence of alarms does not prove that deferral is safe or cost-free.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. CMMS and the asset owner hold deferred tasks, due dates and condition; competent duty owners hold mandatory status and risk; Finance holds backlog cost and provisions; service owners hold consequence; Procurement holds parts and market lead times. Competent risk and statutory decisions prevail over cost estimates and telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat tasks without current owner, changed scope, stale condition, missing consequence, open work not reconciled to backlog or cost estimates outside the current market as uncertain. Show conflicts side by side without silent reconciliation and route them to Accountable asset and duty owner, Finance, service owner and Procurement.

ANALYSIS REQUIRED

Operation type. Lookup, estimate and recommendation limited to attention priority. **Decision interval.** Monthly for material backlog and at budget planning. **Minimum evidence.** Current deferred-task register, mandatory status, condition, service consequence, age, feasible action, cost range, lead time and owner risk assessment. **Method.** Separate mandatory and safety-critical items; estimate incremental cost and service exposure under defer, fund and fail scenarios; avoid adding risks as if independent where they share dependencies. **Uncertainty.** Show backlog GBP range, age, consequence band, evidence confidence, correlated exposure and amount with no current condition evidence. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If competent risk or condition evidence is missing, escalate for owner assessment and do not assign a low-risk status.

ANSWER BOUNDARY

Do not certify deferral as safe, change maintenance priority, authorise spending or translate a financial estimate into a safety decision. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can distinguish manageable deferral from backlog creating material cost, continuity or duty-holder exposure.

FP-048

STAKEHOLDER QUESTION

Is targeted refurbishment, further condition investigation or replacement the most proportionate risk-adjusted option for this asset and its dependencies?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed condition, performance, energy, alarm and run-hour telemetry may establish the asset baseline and verify selected post-work outcomes. **Coverage, timing and aggregation.** Coverage must include the serialised asset, critical components, interfaces and representative operating states; hidden condition and external obsolescence remain unknown. Use aligned pre-intervention and representative post-intervention periods at native cadence, retaining changed controls and demand. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot establish hidden condition, statutory suitability, supplier workmanship, achievable life extension or future support.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Competent engineer and asset owner hold condition, feasible scope and life expectation; Procurement holds supplier, warranty and market evidence; Finance owns cost horizon; service owner holds downtime and performance need; IT or Estates holds dependencies. Competent technical feasibility and mandatory compliance gates prevail over cost comparison. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat intrusive condition not surveyed, interfaces not assessed, obsolescence not confirmed, refurbishment scope without acceptance criteria or replacement costs on a different service basis as incomplete. Show conflicts side by side without silent reconciliation and route them to Competent asset owner, service owner, Procurement and Finance.

ANALYSIS REQUIRED

Operation type. Estimate, comparison and forecast. **Decision interval.** Before option approval and after any material condition finding. **Minimum evidence.** Competent condition evidence, defined refurbishment and replacement scopes, common service outcome, life and failure ranges, downtime, integration, energy, support and end-of-life costs. **Method.** Model both options over the same horizon with residual life, repeat intervention and failure scenarios; include option value of staged replacement where relevant. **Uncertainty.** Show risk-adjusted cost range, life-extension range, break-even failure rate, downtime exposure and sensitivity to hidden condition and support horizon. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If technical feasibility or hidden condition is unresolved, present a survey decision rather than an option ranking.

ANSWER BOUNDARY

Do not certify life extension, choose the option, approve intrusive work or represent modelled savings as guaranteed. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can choose the next proportionate step: more condition evidence, targeted refurbishment or replacement planning.

FP-049

STAKEHOLDER QUESTION

How should a multi-year replacement portfolio be sequenced within funding, outage, market-capacity and dependency constraints?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset condition, failure, run-hour, capacity and service-availability telemetry may inform urgency and feasible outage windows for mapped assets. **Coverage, timing and aggregation.** Coverage is limited to instrumented assets and current operating patterns; future demand, shared dependencies, unobserved failure modes and construction interactions remain scenarios. Use monthly health and demand summaries with event drill-down and retain seasonal peaks, closures and configuration changes. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot set capital priority, promise outage access, establish supplier capacity or certify future condition.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Asset and service owners hold criticality, condition and continuity; Finance owns annual funding limits and accounting treatment; Procurement holds market lead times and packaging options; project and Estates owners hold dependencies, access and delivery capacity. Competent risk decisions and approved funding constraints prevail over model sequencing. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat stale condition, unknown dependencies, unfunded enabling works, market capacity not tested, outage windows not owner-confirmed or shared resources double-booked as uncertainties. Show conflicts side by side without silent reconciliation and route them to Portfolio owner, Finance, Procurement, service owners and competent asset or project owners.

ANALYSIS REQUIRED

Operation type. Forecast, calculation and recommendation limited to phasing scenarios. **Decision interval.** Annually with quarterly refresh for material changes. **Minimum evidence.** Complete candidate portfolio, condition and consequence ranges, dependencies, service capacity, funding envelope, procurement lead times, outage constraints and programme resources. **Method.** Optimise or heuristically sequence within hard safety, continuity and annual funding constraints; model dependency clusters and alternative packages; test scenario robustness rather than one exact plan. **Uncertainty.** Show annual GBP range, assets deferred, risk exposure by year, schedule confidence, binding constraints and sensitivity to supplier lead time and failures. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If critical dependencies or competent condition evidence are missing, provide feasible partial scenarios and an explicit evidence plan, not a single sequence.

ANSWER BOUNDARY

Do not approve capital, commit future budgets, reserve outages, certify asset priority or hide assets excluded because evidence is weak. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Leaders can see a deliverable funding profile, the cost of deferral and where outage, market or dependency evidence changes sequence.

FP-050

STAKEHOLDER QUESTION

After an asset change, which expected cost, energy, reliability and service benefits were realised, uncertain or displaced elsewhere?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed energy, run-hour, alarm, condition, service-availability and environmental telemetry may measure agreed outcomes for the changed asset and affected service zones. **Coverage, timing and aggregation.** Coverage must include an aligned baseline, commissioned post-change points and relevant upstream or downstream systems; changed demand, weather and operating rules remain confounders. Use native intervals and compare representative pre and post periods with matching season, schedule and load where possible. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot prove sole causation, user benefit, whole-life value or absence of displaced cost outside coverage.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Project and asset owners hold approved benefit definitions, baseline and acceptance; Finance owns realised cost; Energy owns utility evidence; CMMS owns maintenance and reliability; service owners hold continuity outcomes; Procurement holds supplier commitments. Pre-agreed benefit definitions and authoritative source records prevail over retrospective claims. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing baseline, changed scope, incomplete commissioning, short observation, tariff changes, cost transferred to another budget or measures selected after results as limitations. Show conflicts side by side without silent reconciliation and route them to Project benefits owner, Finance, asset and service owners, Energy and Procurement.

ANALYSIS REQUIRED

Operation type. Observation, calculation and comparison. **Decision interval.** At commissioning, agreed seasonal checkpoints and 12 months where feasible. **Minimum evidence.** Pre-agreed measures and baseline, complete capital and operating cost, matched pre and post periods, commissioning status, maintenance and service outcomes, and known external changes. **Method.** Use normalised before-after comparison and, where available, a suitable control or counterfactual; reconcile gross, net and displaced effects and test persistence. **Uncertainty.** Show benefit range, confidence interval or scenario band, coverage, payback movement, displaced GBP or carbon and unresolved attribution. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If baseline, observation length or confounding is inadequate, report observed change only and no realised-benefit claim.

ANSWER BOUNDARY

Do not certify value, guarantee persistence, attribute every change to the project, approve final benefits or conceal adverse and displaced effects. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance and asset owners can update future business-case assumptions using measured outcomes and visible displaced effects.

FP-051

STAKEHOLDER QUESTION

Which tariff periods, mapped loads and verified operating events account for the largest current electricity, gas, heat and water costs?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed main and submeter electricity, gas, heat and water readings, BMS plant states and asset power or run-hour telemetry may disaggregate cost drivers. **Coverage, timing and aggregation.** Coverage must follow the current meter hierarchy and mapped service areas; shared plant, landlord supplies, unmetered loads and meter changes remain explicit residuals. Keep revenue or source meter intervals, commonly 15-30 minutes, align tariff boundaries and aggregate to day and billing month. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Meter correlation with plant state does not prove avoidable consumption, fault, ownership or causal responsibility.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Energy or Estates owns meter hierarchy, reads and operating records; Finance owns posted utility cost; Procurement owns effective supply contracts; suppliers or market operators own tariffs and settlement adjustments; service owners hold operating events. Accepted billing meters, effective tariffs and current owner mappings prevail over inferred submeter allocations. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat estimated reads, meter exchanges, missing multiplier, unaligned tariff clock, less than 95% interval completeness, unposted credits or event records not effective for the period as incomplete. Show conflicts side by side without silent reconciliation and route them to Energy manager, Finance, Procurement utility contract owner and relevant service owner.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and diagnosis. **Decision interval.** Monthly with weekly exception checks for material loads. **Minimum evidence.** Reconciled billing quantity and cost, effective tariff components, current meter hierarchy, at least 95% interval coverage for time-of-use analysis, operating events and known adjustments. **Method.** Reprice intervals under the effective tariff, allocate directly mapped loads, correlate verified events only within their service boundary and retain unexplained residual. **Uncertainty.** Show GBP and consumption by tariff period and load class, residual percentage, data coverage, price-volume interaction and sensitivity to estimated reads. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If billing meter, tariff or hierarchy cannot be reconciled, report the verified total only and suppress load attribution.

ANSWER BOUNDARY

Do not infer waste, occupant behaviour or asset fault from consumption alone, allocate cost without policy, or direct operating changes. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance and Energy can target tariff, scheduling, load or metering action at the largest verified cost drivers.

FP-052

STAKEHOLDER QUESTION

After weather, schedule and service conditions are normalised, is heating and cooling cost within the credible range for the covered zones?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed heat, gas and electricity meters, outdoor temperature, zone temperature, valve or damper state, plant run-state and flow or return-temperature points may support weather and schedule normalisation. **Coverage, timing and aggregation.** Coverage must map plant to served zones and operating schedules; shared systems, open windows, local equipment, thermal storage and unmeasured zones remain limitations. Use 5-15 minute BMS data and accepted weather observations, aggregated to occupied and unoccupied degree-hour periods. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Temperature and run-state do not prove comfort, control quality, waste or correct system design.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Energy and Estates own meters, BMS points, schedules and plant mapping; Finance owns cost; Procurement holds tariffs; service owners hold required hours and conditions; competent engineers own control interpretation. Current service requirements and competent system mapping prevail over historical room use and telemetry assumptions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat sensor bias, missing schedule version, changed zone use, weather station mismatch, manual overrides, low coverage or plant-to-zone mapping not verified as confounders. Show conflicts side by side without silent reconciliation and route them to Energy manager, Estates controls owner, service owner, Finance and competent engineer.

ANALYSIS REQUIRED

Operation type. Observation, calculation and comparison. **Decision interval.** Monthly and across representative heating and cooling seasons. **Minimum evidence.** Current schedule and service conditions, mapped meters and zones, accepted weather, at least 90% aligned intervals, tariff, override history and known building changes. **Method.** Normalise energy and cost by degree hours, occupied hours and service output; compare like weather and schedule periods and inspect residuals for documented events. **Uncertainty.** Show normalised cost range, model error, coverage, occupied and unoccupied contributions, residual patterns and sensitivity to base temperature and schedule assumptions. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If mapping, weather or schedule evidence is inadequate, report raw cost and the missing normalisation basis only.

ANSWER BOUNDARY

Do not declare discomfort, poor control or avoidable energy from model residuals alone, change setpoints or schedules, or certify system efficiency. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can separate weather and required service from abnormal cost and decide where competent controls investigation merits funding.

FP-053

STAKEHOLDER QUESTION

What share of utility cost is a verified out-of-hours baseload, and which mapped systems explain the largest supported portion?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed main and submeter electricity, heat and water data, BMS schedules, plant states, server-room loads and asset run-hours may corroborate out-of-hours demand. **Coverage, timing and aggregation.** Coverage is limited to the current meter hierarchy and verified operating-hour definitions; essential continuous loads, cleaning, research, security and delayed shutdowns must remain separate. Use 15-minute intervals and aggregate by approved operating, transition and closed windows for representative weeks. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** A persistent load does not establish unnecessary use, ownership or permission to switch equipment off.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Energy owns meter data and hierarchy; Estates and service owners own schedules and essential-load classification; IT owns server and network service needs; Finance owns cost; Procurement holds tariff terms. Current approved operating requirements and safety or continuity duties prevail over low-load heuristics. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat stale schedules, changed opening, incomplete submeters, clock drift across BMS and meters, temporary projects or unclassified continuous loads as unresolved. Show conflicts side by side without silent reconciliation and route them to Energy manager, Estates, IT where relevant, service owners and Finance.

ANALYSIS REQUIRED

Operation type. Observation, calculation and diagnosis. **Decision interval.** Monthly, with seasonal representative-week comparison. **Minimum evidence.** Approved operating windows, reconciled interval meters, verified load mapping, essential-load register, at least 90% coverage and documented exceptions. **Method.** Estimate the lower envelope within genuine closed periods, allocate only directly metered or strongly time-linked loads, and retain an unexplained residual without claiming causation. **Uncertainty.** Show baseload kW and GBP range, percentage directly metered, residual, weekly variability, coverage and sensitivity to the closed-window definition. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If closed periods or essential loads are not authoritative, do not label consumption out of hours; report time profiles only.

ANSWER BOUNDARY

Do not identify researchers or staff, infer unauthorised activity, switch systems, label essential load waste or allocate cost without owner policy. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Energy and Finance can size the supported baseload opportunity and target meter or owner checks before operational action.

FP-054

STAKEHOLDER QUESTION

Does each utility invoice reconcile from accepted reads through effective rates, standing charges, taxes, estimates, credits and adjustments?

SENSORS AND TELEMETRY

Proposed modality and placement. Accepted main billing meters are the relevant physical evidence; proposed submeters can support reasonableness checks but not replace the billing basis unless contractually recognised. **Coverage, timing and aggregation.** Coverage must match meter serial, multiplier, supply point and exact billing dates; estimated reads, exchanges, rollover and shared supplies are explicit. Use native cumulative and interval reads, preserve read type and align opening and closing boundaries. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Submeter totals, BMS points or consumption models do not establish the legally billed quantity.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Energy or Estates owns meter identity and accepted reads; Procurement holds the supply contract and rates; Finance holds invoices, taxes and payments; supplier holds settlement and adjustment records; relevant tax owners hold treatment. Effective contract terms and accepted billing meter records prevail. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat wrong serial or multiplier, estimated read without status, missing adjustment trail, superseded rate, tax inconsistency, meter exchange not bridged or unmatched supply point as unresolved. Show conflicts side by side without silent reconciliation and route them to Finance, Energy, Procurement utility contract owner and supplier manager.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and comparison. **Decision interval.** Per invoice before payment action and monthly reconciliation. **Minimum evidence.** Meter identity and accepted reads, contract rate schedule, tariff periods, standing and pass-through charges, taxes, prior estimates, credits and meter-change records. **Method.** Reconstruct billed quantity and price from opening to closing read; reprice each component; bridge supplier invoice, calculated amount and posted cost line by line. **Uncertainty.** Show GBP and quantity variance, tolerance, estimated-read exposure, unresolved adjustments and sensitivity to missing interval allocation. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If billing meter identity or effective rate is not verified, raise a specific evidence query and do not state a correct payable amount.

ANSWER BOUNDARY

Do not alter meter records, approve or withhold payment, interpret disputed supply terms or expose restricted tariff data outside the authorised team. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can challenge exact billing differences while preserving a reproducible route from accepted read to invoice and ledger.

FP-055

STAKEHOLDER QUESTION

What cost and operational-carbon range is associated with a proposed schedule, setpoint or control-sequence change within the approved service envelope?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed energy meters, plant run-state, temperatures, pressures, flow, valve or damper positions, CO2 and occupancy proxies may establish the current control-response baseline. **Coverage, timing and aggregation.** Coverage must include the affected plant, served zones, required service periods and relevant weather; shared systems, local overrides and unmeasured comfort effects remain limitations. Use 5-15 minute aligned intervals across representative baseline and trial periods and retain all setpoint and override changes. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Telemetry does not authorise a control change, establish safety or comfort adequacy, or guarantee savings.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates controls and competent engineering owners hold sequences, limits and safety; service and accessibility owners hold required conditions; Energy owns meters and carbon calculations; Finance owns cost; Procurement holds tariffs and maintenance terms. Competent operating limits and current service requirements prevail over cost or carbon estimates. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat uncommissioned points, unstable baseline, changed occupancy or weather not modelled, manual overrides, unmet service conditions or carbon factors not effective for the period as confounders. Show conflicts side by side without silent reconciliation and route them to Competent controls owner, service owner, Energy, Finance and Estates.

ANALYSIS REQUIRED

Operation type. Estimate, forecast and comparison. **Decision interval.** Before any authorised trial and after a representative monitored trial. **Minimum evidence.** Approved operating envelope, mapped healthy telemetry, representative baseline, proposed change, tariff and carbon factors, service constraints and trial acceptance criteria. **Method.** Build a physics-informed or transparent statistical baseline; simulate candidate settings within hard constraints; if trialled, compare matched periods and check rebound or displaced load. **Uncertainty.** Show annualised GBP and carbon range, model error, service-condition exceedance, coverage and sensitivity to weather, demand and control response. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the service envelope or baseline is not reliable, offer no change recommendation and identify the commissioning or measurement need.

ANSWER BOUNDARY

Do not change controls, certify comfort, air quality, safety or savings, or optimise cost by weakening required service conditions. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Owners can decide whether a controlled competent trial is worth undertaking and define the evidence that would accept or reject it.

FP-056

STAKEHOLDER QUESTION

What is the cost of meeting current ventilation and indoor-environment requirements, and where is an evidence-backed efficiency opportunity?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed airflow, fan power, pressure, damper state, CO2, temperature, humidity and occupancy-proxy measurements may quantify ventilation service for mapped air-handling systems and zones. **Coverage, timing and aggregation.** Coverage must include supply and extract paths, served zones and representative operating states; sensor placement, mixing, local sources and natural ventilation create spatial limits. Use 1-5 minute control data aggregated to occupied-hour distributions and exceedance duration, not single averages. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** CO2 is not a direct universal airflow measure, and sensor values cannot certify indoor-air quality, infection risk or regulatory compliance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Competent ventilation and duty owners hold required airflow and control limits; Estates owns plant and BMS mapping; service and accessibility owners hold occupancy and use requirements; Energy and Finance own cost evidence; Procurement holds tariffs and contracts. Competent requirements and current commissioning records prevail over inferred demand. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat uncalibrated CO2 or airflow, unknown occupancy basis, changed room use, sensor placement not validated, simultaneous heating or cooling effects, and uncommissioned controls as limitations. Show conflicts side by side without silent reconciliation and route them to Competent ventilation owner, Estates, service owner, Energy and Finance.

ANALYSIS REQUIRED

Operation type. Observation, calculation and diagnosis. **Decision interval.** Monthly and during representative occupied seasons. **Minimum evidence.** Current ventilation requirements, commissioning data, mapped healthy sensors, representative operating hours, fan and conditioning energy, tariff and service-condition evidence. **Method.** Calculate fan and conditioning cost by operating state; compare delivered service with required ranges; identify simultaneous or excessive operation only where airflow and control sequence are verified. **Uncertainty.** Show GBP per occupied hour or delivered-air unit where valid, coverage, measurement error, exceedance duration and savings range subject to unchanged service. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If required airflow, occupancy basis or sensor validity is unknown, report energy and condition observations separately and no efficiency claim.

ANSWER BOUNDARY

Do not reduce ventilation, change controls, certify indoor-air conditions or infer room occupancy from CO2 alone. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Energy and Estates can target commissioning or design investigation where cost is high and service evidence supports an opportunity.

FP-057

STAKEHOLDER QUESTION

Is the water-cost increase better explained by physical loss, changed demand, meter quality or billing treatment, and what should be checked first?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed main and submeter flow, pulse counts, pressure, tank level and leak-detection sensors may corroborate water behaviour across mapped supply branches. **Coverage, timing and aggregation.** Coverage must follow the current meter and pipe hierarchy; shared branches, intermittent uses, fire systems, flushing and unmetered outlets remain explicit. Use 1-15 minute flow and pressure data, aggregate to night-flow and daily totals, and retain zero-flow and sensor-reset periods. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Continuous flow does not prove a leak, location, responsibility or safe isolation point.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Energy or Estates owns meter and pipe mapping, operating events and competent findings; Finance holds cost and invoice; Procurement holds tariff; service owners hold changed demand; water supplier holds billing adjustments. Accepted billing reads and competent leak or meter findings prevail over anomaly scores. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat estimated bills, meter exchange, missing pulse factor, low pressure events, planned flushing, changed opening, sensor drift or hierarchy not verified as competing explanations. Show conflicts side by side without silent reconciliation and route them to Estates water owner, Energy, Finance, Procurement utility owner and affected service owner.

ANALYSIS REQUIRED

Operation type. Diagnosis, observation and calculation. **Decision interval.** Daily for material anomalies and per billing period. **Minimum evidence.** Reconciled billing quantity, verified meter hierarchy, at least 14 comparable days including closed periods where available, operating events, pressure context and current tariff. **Method.** Compare night flow, daily profile and branch balance with baseline and known demand; test meter and billing explanations before localising a suspected loss. **Uncertainty.** Show excess-volume and GBP range, branch-balance residual, detection confidence, coverage and the observations supporting each competing explanation. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If hierarchy, baseline or meter health is inadequate, state abnormal consumption only and route a competent meter or leak check.

ANSWER BOUNDARY

Do not declare a leak, direct isolation, certify water safety, attribute fault or enter restricted areas. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can route the issue first to billing, metering or physical investigation and prioritise the branch with strongest evidence.

FP-058

STAKEHOLDER QUESTION

What payback, net-cost and delivery-risk range is supportable for the proposed energy or water efficiency investment?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed accepted meters and relevant plant, environmental and service telemetry may establish the baseline and expected operating response for the project boundary. **Coverage, timing and aggregation.** Coverage must include all materially affected energy or water flows, service conditions and representative seasons; interactive effects, rebound and displaced loads remain scenarios. Use native intervals, at least one representative seasonal cycle where material, and an agreed baseline model with retained raw lineage. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot establish capital price, compliance, future maintenance, user benefit or guaranteed persistence.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Project and Energy owners hold scope, baseline and technical assumptions; Finance owns discount, inflation and affordability; Procurement holds market prices and supplier terms; service and competent owners hold operating constraints; sustainability owner holds carbon factors. Authoritative scope, competent feasibility and current commercial evidence prevail over generic benchmark savings. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat short or unrepresentative baseline, unpriced enabling works, supplier savings without method, expired quotes, missing maintenance effect or carbon factors not effective for the scenario as incomplete. Show conflicts side by side without silent reconciliation and route them to Project owner, Energy, Finance, Procurement and competent service owner.

ANALYSIS REQUIRED

Operation type. Calculation, estimate and forecast. **Decision interval.** At business-case formation and after priced offers. **Minimum evidence.** Competent feasible scope, representative baseline, tariff and carbon factors, capital and recurring cost, delivery schedule, persistence, maintenance, service constraints and uncertainty ranges. **Method.** Normalise baseline, model gross and net savings including interactions and degradation, discount cash flows, and run scenario or probabilistic analysis for price, performance and delivery risk. **Uncertainty.** Show simple and discounted payback ranges, net present cost or value, P10-P50-P90 benefit where supportable, carbon range and break-even assumptions. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If baseline, feasibility or cost perimeter is incomplete, provide a value-of-information case and no payback claim.

ANSWER BOUNDARY

Do not certify savings or carbon, approve the project, select a supplier, set discount policy or treat modelled payback as guaranteed. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Decision-makers can compare the project with other calls on capital and see which uncertain inputs most affect the case.

FP-059

STAKEHOLDER QUESTION

Which allocation method makes shared utility cost most transparent when direct metering is incomplete, and how uncertain are the charges?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed main meters, available submeters, plant run-hours, thermal or electrical load proxies, opening hours and anonymous zone counts may support a bounded allocation. **Coverage, timing and aggregation.** Coverage must follow current service and meter mappings; unmetered common plant, circulation, losses and overlapping zones form a visible residual pool. Use aligned interval data and aggregate to accounting month after direct-meter reconciliation. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Proxy activity does not establish ownership, benefit, fairness or actual consumption by a cost centre.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns allocation policy; Energy owns utility totals, meter hierarchy and loss treatment; Estates owns current service areas and space; service owners confirm operational drivers; Procurement holds tariff terms. Reconciled billing total and approved policy prevail; current owner records override the May 2021 floor plan. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unverified service mapping, changed room ownership, less than 90% relevant coverage, overlapping meters, unexplained losses or policy not effective for the period as incomplete. Show conflicts side by side without silent reconciliation and route them to Finance, Energy, Estates and affected budget holders.

ANALYSIS REQUIRED

Operation type. Calculation, estimate and comparison. **Decision interval.** Monthly with annual policy confirmation. **Minimum evidence.** Reconciled utility total, approved allocation hierarchy, direct meter coverage, current service areas, defensible proxy drivers and agreed treatment of common losses. **Method.** Allocate direct reads first, then model the shared pool with transparent drivers; preserve a residual and compare at least two plausible allocations without implying one is measured truth. **Uncertainty.** Show GBP by cost centre as a range, directly metered percentage, proxy share, residual, coverage and sensitivity to each driver. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If total, policy or mapping is not verified, withhold cost-centre allocation and report only the aggregate evidence gap.

ANSWER BOUNDARY

Do not track individuals, infer attendance, create allocation policy, treat proxy output as a bill or expose another cost centre's restricted data. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can allocate with visible uncertainty and invest in submetering only where it would materially change charges.

FP-060

STAKEHOLDER QUESTION

Can every reported utility-cost and operational-carbon total be reproduced end to end from accepted source evidence?

SENSORS AND TELEMETRY

Proposed modality and placement. Accepted billing meters and verified submeters provide the physical quantity evidence; BMS points may support checks but are not substitutes for the declared reporting boundary. **Coverage, timing and aggregation.** Coverage must identify each supply point, meter hierarchy, excluded source and reporting period; meter changes and estimated reads require explicit bridges. Use source-native readings and preserve each aggregation, unit conversion and time-zone treatment. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Meter completeness alone does not validate tariff, carbon factor, organisational boundary or renewable accounting treatment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Energy owns meter data, hierarchy and energy totals; Finance owns posted cost; Procurement owns effective tariffs; sustainability owner holds reporting boundary and carbon factors; suppliers hold invoices and adjustments. Declared reporting rules, accepted meter reads and effective factors prevail over dashboard defaults. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unknown factor version, mixed location-based and market-based treatment, missing unit conversion, unreconciled invoice, estimated read, absent source lineage or material unexplained residual as defects. Show conflicts side by side without silent reconciliation and route them to Energy, Finance, Procurement and sustainability reporting owner.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and comparison. **Decision interval.** Monthly and at each formal reporting cut-off. **Minimum evidence.** Complete source list, accepted reads or invoices, meter hierarchy, effective tariffs and carbon factors, unit conversions, adjustments, reporting boundary and transformation lineage. **Method.** Recompute each total independently from source; reconcile quantity, cost and carbon bridges; test control totals and disclose every manual adjustment. **Uncertainty.** Show reproducibility percentage, unexplained quantity and GBP, carbon-factor sensitivity, source coverage and rounding tolerance. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material source, factor or bridge is missing, mark the affected total not reproducible and suppress any precise derived claim.

ANSWER BOUNDARY

Do not certify financial or carbon reporting, choose accounting treatments, conceal manual adjustments or disclose restricted utility terms. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can defend reported totals, locate the exact break in lineage and avoid reliance on irreproducible dashboards.

FP-061

STAKEHOLDER QUESTION

Before the commitment reaches its next authority point, which mandatory finance, procurement, legal, technical and owner evidence is present or missing?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor establishes governance readiness. Optional site surveys, commissioning evidence or measured baselines may support a technical annex only where the required authority explicitly calls for them. **Coverage, timing and aggregation.** Any measurement is limited to the named scope, asset, location and validity period and must be linked to its competent owner. Use the stated survey or test cadence and preserve the evidence snapshot submitted at the authority point. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot substitute for budget authority, competition, delegated approval, legal advice or competent sign-off.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns budget and delegated financial authority; Procurement owns route and commercial record; Legal owns legal interpretation; project, service, technical, safety, accessibility and data owners hold their required evidence. Current policy, delegated authority and owner decisions prevail over checklist completion or historical approvals. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat expired approvals, conditions not discharged, scope or value changed since authority, missing conflict declarations, unsigned owner acceptance or evidence retrieved after the decision without effective dating as blockers. Show conflicts side by side without silent reconciliation and route them to Named decision secretariat, Finance, Procurement, Legal and each accountable owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to evidence actions. **Decision interval.** At every defined authority point and after any material scope, value or risk change. **Minimum evidence.** Current authority matrix, complete proposal scope and value, required evidence list, owner decisions, conditions, conflicts and effective policy versions. **Method.** Build a requirement-to-evidence matrix; classify present, valid, conditional, expired, conflicted or missing; verify that evidence matches the current proposal rather than an earlier version. **Uncertainty.** Show completeness by mandatory domain, material conditions, age, value changed since authority and unresolved owner decisions. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If any mandatory authority or evidence item is absent, state not ready for the next authority point and identify the owner action.

ANSWER BOUNDARY

Do not grant authority, waive a condition, interpret silence as consent or present checklist completeness as approval or compliance. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can stop avoidable late-stage failure and route each missing requirement to the authority that owns it.

FP-062

STAKEHOLDER QUESTION

Are all supplier credentials, insurances, licences, memberships and competence records current, scope-specific and independently verifiable?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor verifies supplier credentials or competence. Site access or training-system events may show that a named prerequisite was recorded before attendance, but they are not proof of validity or capability. **Coverage, timing and aggregation.** Coverage is limited to the authorised contractor, role, site and attendance event; subcontractors, scope changes and off-site activity require separate evidence. Use event timestamps and retain the credential reference and validity status used at the time. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Access granted, a badge scan or training completion does not certify competence, insurance, licence scope or ongoing validity.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the contract owner hold required supplier evidence and verification record; Legal or Risk owners hold insurance requirements; competent duty holders define licences and competence; issuing bodies own credential status; suppliers provide source documents. Independent issuing-body status and current owner requirements prevail over supplier self-declaration. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat expired documents, unverifiable issuers, scope mismatch, missing subcontractor coverage, changed personnel or insurance limits below current requirement as invalid or unresolved. Show conflicts side by side without silent reconciliation and route them to Procurement supplier owner, Risk or Legal, competent duty owner and accountable contract owner.

ANALYSIS REQUIRED

Operation type. Lookup and comparison. **Decision interval.** Before award, before first attendance and at each expiry or scope change. **Minimum evidence.** Scope-specific requirement list, supplier and subcontractor identities, original credential references, issuing-body verification, validity dates, insurance limits and named competent-owner acceptance. **Method.** Match each requirement to independently verified evidence and applicable entity or role; classify current, expiring, not applicable, insufficient or unverified. **Uncertainty.** Show coverage by mandatory requirement, days to expiry, unverified entities and value or work scope affected without scoring competence beyond evidence. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a mandatory credential cannot be verified, state that readiness is not established and route the case to the owner.

ANSWER BOUNDARY

Do not certify competence, approve a supplier, grant site access, interpret legal sufficiency or disclose personal credential details beyond authorised need. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can prevent mobilisation with expired or inapplicable evidence and request the exact scope-specific confirmation required.

FP-063

STAKEHOLDER QUESTION

Are sustainability, labour, community and social-value promises converted into measurable deliverables with owners, baselines, dates and evidence?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor validates a social-value promise. Optional energy, waste, travel, material or attendance counts may measure a defined deliverable only where privacy, attribution and boundary rules are agreed in advance. **Coverage, timing and aggregation.** Coverage must match the contracted activity, beneficiary group, location and reporting period; wider organisational activity and existing provision remain outside. Use the contractually stated cadence and aggregate to the least granular level needed. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** A count does not establish quality, additionality, beneficiary experience, fair work or supplier contribution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the contract owner hold tender commitments and contract deliverables; sustainability, HR, community, equality and service owners define valid measures; suppliers provide evidence; Finance holds any linked price or remedy. Effective contractual wording and authoritative owner definitions prevail over marketing statements. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat vague promises, no baseline, no beneficiary or owner, activity outside the contract, double counting, self-reported totals without source or measures added after award as unsupported. Show conflicts side by side without silent reconciliation and route them to Procurement contract owner and the relevant sustainability, people, community or equality owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to measurable contract wording. **Decision interval.** Before award and at each scheduled delivery checkpoint. **Minimum evidence.** Each promised outcome, baseline, unit, target, date, beneficiary, evidence source, attribution rule, owner, data permission and consequence of non-delivery. **Method.** Convert promises into a deliverable matrix; separate input, activity, output and outcome; test measurability, additionality and non-duplication; retain qualitative evidence where numbers are inappropriate. **Uncertainty.** Show percentage of commitments with complete measures, unsupported targets, attribution confidence and the value linked to unresolved deliverables. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a promise cannot be measured or owned proportionately, label it narrative and do not create a false numeric target.

ANSWER BOUNDARY

Do not certify social value, identify beneficiaries, weaken labour or equality duties, score a supplier beyond approved criteria or promise a contract remedy. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can contract for measurable, owned evidence rather than broad promises that cannot be managed or valued.

FP-064

STAKEHOLDER QUESTION

Are conflicts of interest, related-party links and segregation-of-duty controls complete for each procurement, approval and payment role?

SENSORS AND TELEMETRY

Proposed modality and placement. No building sensor is relevant. Identity, relationship and approval evidence must come from authorised governance systems and declarations, not location, access or communication surveillance. **Coverage, timing and aggregation.** No physical or behavioural telemetry should be collected for this purpose. Use only authorised record timestamps and the minimum role-level information needed. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Co-location, common access times or communication patterns must never be treated as evidence of a conflict or collusion.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement, Finance, HR or governance owners hold declarations, roles, delegations and approval records; Legal or Ethics owners interpret conflicts; corporate registries may provide permitted independent entity evidence. Current declarations, authoritative role records and formal owner determinations prevail over inferred relationships. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat missing declarations, outdated role assignments, shared accounts, approval after transaction, entity-name mismatch or a related-party question awaiting owner determination as unresolved. Show conflicts side by side without silent reconciliation and route them to Governance or Ethics owner, Procurement, Finance and Legal.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to governance routing. **Decision interval.** Before evaluation, award, order approval, receipt and payment where required. **Minimum evidence.** Current participants and roles, required declarations, supplier entity and ownership evidence, delegation matrix, approval sequence and authorised conflict determination. **Method.** Apply deterministic segregation rules and entity matching; flag missing or incompatible roles; separate factual relationship evidence from owner judgement. **Uncertainty.** Show control coverage, missing declarations, role conflicts, entity-match confidence and transactions affected, without producing a misconduct score. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If identity or relationship evidence is ambiguous, pause the analytic conclusion and route confidentially to the authorised owner.

ANSWER BOUNDARY

Do not infer personal relationships, monitor communications, declare misconduct, expose declarations beyond authorised roles or replace an Ethics or Legal determination. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can correct role conflicts and missing declarations before procurement, approval or payment authority is exercised.

FP-065

STAKEHOLDER QUESTION

Where does the proposed procurement create single-supplier, proprietary, geographic, data or operational dependency, and what exposure follows?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed interface inventories, protocol and version data, asset dependency telemetry, service availability and stock or lead-time signals may corroborate operational coupling. **Coverage, timing and aggregation.** Coverage is limited to documented technical and physical dependencies; supplier finances, legal rights, market substitutes and uninstrumented manual work remain outside. Use current configuration snapshots with monthly service summaries and retain effective version dates. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Installed dependence does not prove unavoidable lock-in, supplier risk or absence of a feasible migration path.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns supplier and market evidence; technical and service owners hold architecture, dependency and continuity needs; Legal holds licence, data and termination terms; Finance holds cost exposure; suppliers provide support and roadmap evidence. Current architecture, effective rights and independently tested market evidence prevail over vendor assertions or telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat undocumented interfaces, expiring support, data export not tested, substitute capacity not current, concentration hidden across subcontractors or migration cost omitted as unresolved. Show conflicts side by side without silent reconciliation and route them to Procurement, Legal, competent technical owner, service owner and Finance.

ANALYSIS REQUIRED

Operation type. Diagnosis, comparison and recommendation limited to dependency treatments. **Decision interval.** At procurement strategy, evaluation and before renewal. **Minimum evidence.** Current dependency map, mandatory outcomes, interface and data portability evidence, supplier concentration, alternatives, transition time and cost, continuity requirements and contract exit rights. **Method.** Map single points of failure and control; compare open, multi-source, escrow, stock, option and transition treatments; quantify dependency after mitigations rather than counting suppliers alone. **Uncertainty.** Show concentration metrics, transition cost and time ranges, untested interfaces, residual continuity exposure and market-evidence age. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If data portability, substitute capability or legal rights are unverified, state dependency unresolved and require a test or owner decision.

ANSWER BOUNDARY

Do not label a supplier unsafe or insolvent, mandate a technology, select a mitigation, inspect proprietary content or certify resilience. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can price dependency and include proportionate exit, interoperability, data-portability or continuity provisions.

FP-066

STAKEHOLDER QUESTION

Are privacy, cyber-security, data-governance, retention and access requirements proportionate, testable and assigned across the contract?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed device inventory, firmware version, security-status metadata, network-segment membership and access-log aggregates may evidence the deployed boundary without inspecting payloads, speech, research content or individual behaviour. **Coverage, timing and aggregation.** Coverage is limited to authorised assets and interfaces; supplier cloud, subcontractors, shadow devices and off-network paths require separate evidence. Use configuration snapshots and security events at the minimum aggregation needed, with strict role access and retention. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Technical metadata cannot certify security, privacy, legal basis, human impact or supplier assurance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Information security, data protection and system owners hold current requirements and risk decisions; Procurement owns tender and contract evidence; Legal owns terms; service owner defines purpose and minimum data; suppliers provide architecture and assurance material. Current owner-approved requirements and formal risk decisions prevail over generic checklists or vendor claims. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat undefined purpose, excessive data fields, unclear controller roles, unsupported data location, untested deletion or export, end-of-support components or access rights not mapped to roles as blockers. Show conflicts side by side without silent reconciliation and route them to Information security owner, Data Protection owner, system owner, Procurement and Legal.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to requirement and evidence gaps. **Decision interval.** Before tender issue, award, data connection and material change. **Minimum evidence.** Defined purpose and data flow, asset and interface boundary, threat and privacy assessment, role-based access, retention, incident, deletion, portability, support and testable acceptance evidence. **Method.** Map each risk and duty to requirement, supplier response, test and owner; check proportionality and data minimisation; preserve unresolved residual risk for accountable acceptance. **Uncertainty.** Show mandatory-control coverage, untested claims, data-field reduction, residual risk ratings from owners and evidence expiry dates. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If purpose, data flow or accountable risk owner is absent, state not ready for connection or commitment and do not create a technical workaround.

ANSWER BOUNDARY

Do not certify security or privacy, inspect content or network payloads, accept risk, grant access, approve a supplier or disclose sensitive architecture. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can require proportionate, testable controls before a connected product creates unmanaged data or security obligations.

FP-067

STAKEHOLDER QUESTION

Are accessibility, inclusive use and reasonable-adjustment requirements preserved through specification, evaluation, delivery and acceptance evidence?**SENSORS AND TELEMETRY**

Proposed modality and placement. No sensor can establish accessibility or inclusion. Competent measurements of dimensions, force, lighting, acoustics or interface response may support a defined acceptance test, supplemented by authorised user evidence. **Coverage, timing and aggregation.** Coverage is limited to the specified route, product, interface and use cases; diverse needs, temporary barriers and digital or service interactions require separate evidence. Use the stated test conditions and preserve results by requirement without identifying individuals. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Average reach, movement, noise or use data must not define a normal user or certify inclusive experience.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Accessibility and equality owners hold current requirements and adjustment routes; service users or authorised representatives provide needs evidence; technical and Estates owners hold site interfaces; Procurement owns tender and evaluation record; Legal holds duties. Current legal duties, competent accessibility advice and validated user requirements prevail over generic standards or telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat requirements copied without context, inaccessible test coverage, changes after evaluation, missing alternative format, unowned adjustments or acceptance that omits the full journey as incomplete. Show conflicts side by side without silent reconciliation and route them to Accessibility owner, service owner, Procurement, Legal and competent technical owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to evidence and acceptance actions. **Decision interval.** At requirements, evaluation, prototype or sample, installation and acceptance checkpoints. **Minimum evidence.** Current accessibility duties, diverse use cases, adjustment needs, measurable criteria, supplier evidence, competent test, permitted user evidence and named owner for each gap. **Method.** Create a requirement-to-test matrix across the whole service journey; test mandatory gates before weighted comparison; record alternative methods and unresolved user impacts. **Uncertainty.** Show requirement coverage, failed or untested criteria, affected use cases, evidence confidence and cost or schedule effect of closing gaps. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If competent evidence or representative use cases are absent, state accessibility not established and require owner action.

ANSWER BOUNDARY

Do not infer disability, identify users, certify accessibility, trade away mandatory needs for price or represent one user's experience as universal. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can keep inclusion requirements visible through commercial decisions and prevent late, costly corrective changes.

FP-068

STAKEHOLDER QUESTION

Are hazardous materials, controlled waste, take-back and end-of-life responsibilities priced, evidenced and assigned before commitment?**SENSORS AND TELEMETRY**

Proposed modality and placement. Proposed environmental or containment monitoring, waste-container fill or weight, asset identification and chain-of-custody scans may corroborate handling events where authorised. **Coverage, timing and aggregation.** Coverage is limited to mapped materials, containers, work areas and licensed transfer points; hidden materials, contamination, composition and off-site treatment require competent evidence. Use event-level custody records and safety-required monitoring cadence, retaining alarms, gaps and instrument checks. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Sensors cannot identify hazardous composition, certify safe handling, classify waste or prove lawful disposal.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Competent safety, environmental and Estates owners hold surveys, classifications, controls and consents; Procurement and Legal hold supplier duties; waste carriers and facilities hold licences and transfer evidence; asset owners hold end-of-life requirements. Current competent classification, statutory documentation and authorised chain of custody prevail over labels or telemetry. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unknown material composition, expired survey, missing waste code, unlicensed carrier, incomplete transfer note, unclear producer duty or changed scope as blockers. Show conflicts side by side without silent reconciliation and route them to Competent safety or environmental owner, Estates, Procurement, Legal and accountable asset owner.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to evidence actions. **Decision interval.** Before tender, mobilisation, transfer and financial close-out. **Minimum evidence.** Current material and waste classification, work scope, control plan, licensed parties, transfer route, end destination, responsibilities, records and cost provision. **Method.** Map each material and waste stream to duty, owner, permitted route and required evidence; reconcile estimated and transferred quantities without inferring composition. **Uncertainty.** Show evidence coverage, quantity range, missing custody links, licence validity and financial exposure for unresolved end-of-life duties. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If classification or licensed route is absent, state that the activity is not evidence-ready and route it to the competent owner.

ANSWER BOUNDARY

Do not classify materials or waste, direct handling, certify safety or lawful disposal, authorise access or approve a supplier. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Procurement can assign and price take-back, controlled-waste and end-of-life duties before they become residual liabilities.

FP-069

STAKEHOLDER QUESTION

What contract-related social-value outcome is additional, measurable and attributable without double counting existing or unrelated activity?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor establishes social-value attribution. Contract-specific counts, resource use or attendance may support a pre-defined output only when privacy, consent and aggregation rules permit. **Coverage, timing and aggregation.** Coverage must match the contract-funded activity, beneficiary boundary and comparison period; wider university activity and supplier corporate totals remain outside. Use the minimum agreed aggregation and preserve baseline and counterfactual source records. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Activity counts do not establish additionality, quality, long-term outcome or causal attribution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and contract owner hold promised outcomes and delivery scope; community, equality, HR or sustainability owners define valid measures; Finance holds contract value; beneficiaries or authorised evaluators may provide permitted outcome evidence; supplier supplies its records. Pre-agreed attribution rules and authoritative owner evidence prevail over supplier marketing claims. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat no baseline, outcome already funded elsewhere, beneficiaries outside scope, duplicate claims across contracts, self-report without source or changed measure after award as unsupported. Show conflicts side by side without silent reconciliation and route them to Contract owner, Procurement and the relevant community, equality, people or sustainability owner.

ANALYSIS REQUIRED

Operation type. Observation, comparison and estimate. **Decision interval.** At baseline, scheduled delivery points and agreed outcome horizon. **Minimum evidence.** Contract commitment, baseline, beneficiary and activity boundary, counterfactual or contribution logic, source evidence, permission, cost and explicit double-count controls. **Method.** Separate output from outcome; compare with baseline or credible contribution evidence; allocate only the portion supportable under the pre-agreed rule and keep shared outcomes unallocated where necessary. **Uncertainty.** Show attributed range, beneficiary coverage, confidence, excluded existing activity, duplicate-claim checks and time horizon. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If additionality or boundary cannot be supported, report delivered activity only and no attributed outcome.

ANSWER BOUNDARY

Do not identify beneficiaries, certify social value, claim causation from participation alone, monetise outcomes without approved method or alter contract scoring. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can report a defensible contract contribution without inflating claims through double counting or unrelated activity.

FP-070

STAKEHOLDER QUESTION

Which responsible-procurement commitments are met, partial, overdue, contradicted or not assessable, and which owner action is due next?

SENSORS AND TELEMETRY

Proposed modality and placement. Use only contract-relevant, privacy-preserving measurements such as energy, waste, delivery mode, material traceability or service-condition evidence where the commitment defines them. **Coverage, timing and aggregation.** Coverage must match the contracted deliverable, location and period; supplier-wide claims, subcontractors and off-site impacts require separate sources. Use the contractual cadence and preserve source-level exceptions, gaps and validity status. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot establish labour practice, ethical conduct, legal compliance or supplier responsibility beyond the measured phenomenon.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and contract owner hold commitments, evidence schedule and remedies; sustainability, equality, HR, safety, data and community owners hold domain interpretations; suppliers and authorised third parties provide evidence; Legal advises on escalation. Effective contract terms and authoritative domain-owner findings prevail over dashboard status. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat overdue evidence, changed scope, unverifiable source, partial subcontractor coverage, metric outside the agreed boundary or owner dispute as unresolved rather than failed by default. Show conflicts side by side without silent reconciliation and route them to Contract owner, Procurement, relevant domain owner and Legal where required.

ANALYSIS REQUIRED

Operation type. Lookup, comparison and recommendation limited to evidence and escalation routing. **Decision interval.** At each contract checkpoint and immediately for material owner-defined triggers. **Minimum evidence.** Complete commitment register, due dates, evidence rules, supplier submissions, owner assessment, contractual consequence and prior actions. **Method.** Compare evidence with each deliverable and trigger; classify met, partial, overdue, contradicted, not applicable or owner decision; prioritise by consequence and notice deadline. **Uncertainty.** Show coverage, overdue items, confidence, value or scope affected, repeated exceptions and time remaining for contractual action. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If evidence or owner interpretation is incomplete, state unresolved and route the exact action without declaring breach.

ANSWER BOUNDARY

Do not certify compliance, determine breach, direct supplier action, apply a remedy, expose confidential workforce data or replace competent-owner judgement. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The contract team can focus on material commitments, preserve deadlines and route the next evidence or owner action precisely.

FP-071

STAKEHOLDER QUESTION

What is the supportable low-base-high or percentile year-end outturn for each material cost centre, contract and project, and what drives the range?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor determines financial outturn. Optional metered consumption, demand counts, work progress or asset events may support specific volume or timing drivers when linked to the forecast line. **Coverage, timing and aggregation.** Coverage is limited to mapped costs, assets and periods; future prices, orders, claims, owner decisions and unconnected commitments remain scenarios. Use source cadence and aggregate to forecast week and accounting month with late events identified. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Telemetry cannot establish accounting recognition, future scope, supplier delivery or available budget.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns actuals, commitments, accounting rules and forecast framework; Procurement owns contract and order terms; project, service, Energy and asset owners hold current cost-to-complete and volume drivers; budget holders own assumptions. Posted actuals, effective commitments and accountable-owner forecasts prevail over statistical extrapolation. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat stale owner forecasts, duplicate commitments, missing accrual basis, unresolved claims, price terms not current or forecast drivers outside their evidence perimeter as uncertainty. Show conflicts side by side without silent reconciliation and route them to Finance business partner, budget holder, Procurement and relevant project or service owners.

ANALYSIS REQUIRED

Operation type. Forecast, calculation and comparison. **Decision interval.** Monthly, increasing to weekly near year-end for material volatility. **Minimum evidence.** Reconciled actuals, open commitments, supported accruals, latest owner forecasts, effective price and scope, schedule, material risks and at least six comparable periods for statistical residuals where used. **Method.** Build line-level remaining-cost forecasts; separate deterministic commitments from scenario items; aggregate with dependence between shared risks and reconcile to prior forecast movement. **Uncertainty.** Show P10, P50 and P90 or bounded low-base-high outturn, forecast error, unresolved GBP, coverage and contribution of each risk driver. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material owner forecast or commitment perimeter is missing, label the affected outturn incomplete and do not hide it in a central estimate.

ANSWER BOUNDARY

Do not approve budget movement, set provisions, promise delivery or represent a percentile as an authorised spending limit. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can challenge the assumptions creating the widest outturn range and manage funding or timing exposure before year-end.

FP-072

STAKEHOLDER QUESTION

How would plausible changes in service demand, opening patterns and asset use alter purchase quantities, utility cost and support capacity?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed anonymous zone or entrance counts, bookings, equipment run-hours, queue measures, utility meters and service-demand events may describe current relationships without identifying people. **Coverage, timing and aggregation.** Coverage must match the affected service, zones, assets and hours; remote activity, latent demand, exceptional events and unobserved user groups remain outside. Use event or 1-15 minute data aggregated to hour, day and representative term or season after clock alignment. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** Observed association does not prove that occupancy causes cost, predict individual attendance or capture policy-driven demand.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Service owners hold scenario definitions and required service levels; Estates holds current opening, capacity and asset mapping; Finance owns cost; Procurement holds price and lead time; Energy owns utility relations; scheduling owners hold planned activity. Accountable scenario assumptions and current owner records prevail over extrapolation and the historical floor plan. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat scenario values outside observed or competent-design ranges, changed room use, sparse demand, unverified asset-to-cost linkage or prices outside their effective period as uncertain. Show conflicts side by side without silent reconciliation and route them to Service owner, Finance, Procurement, Estates and Energy where relevant.

ANALYSIS REQUIRED

Operation type. Estimate, forecast and comparison. **Decision interval.** At budget setting, service-change planning and major purchase decisions. **Minimum evidence.** Clearly defined scenarios, current baseline, representative demand and cost drivers, capacity limits, service-level constraints, price terms and owner-approved relationships. **Method.** Build transparent driver-based scenarios; use elasticities only within validated ranges; enforce capacity and service constraints and compare with a no-change baseline. **Uncertainty.** Show cost and quantity ranges for each scenario, coverage, model error, non-linear thresholds and sensitivity to demand mix and timing. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If relationships are weak or scenarios exceed evidence, present bounded stress cases rather than a predictive forecast.

ANSWER BOUNDARY

Do not identify people, infer individual attendance or productivity, decide opening policy, or treat scenario output as an approved purchase plan. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Budget and procurement teams can test the financial consequence of service choices before fixing quantities or annual envelopes.

FP-073

STAKEHOLDER QUESTION

What contingency or risk allowance is supportable from identified uncertainties without double counting or concealing optimism?

SENSORS AND TELEMETRY

Proposed modality and placement. No sensor sets contingency or accounting provision. Telemetry may support the frequency or consequence distribution of a named operational risk when coverage and failure definitions are valid. **Coverage, timing and aggregation.** Coverage is limited to mapped assets, events and periods; market, legal, scope, inflation and governance risks require other sources. Use event-level history and exposure-normalised aggregates, retaining data gaps and changes in detection. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Observed faults do not establish accounting treatment, future independence of risks or the correct risk appetite.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns contingency policy, provision criteria and accounting treatment; project and risk owners hold risk register, probability and consequence; Procurement holds market and contract risks; competent owners hold technical and safety consequences. Current Finance policy and accountable risk-owner estimates prevail over automated distributions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat generic percentages, duplicated risks, probabilities without basis, risks already embedded in base cost, stale quotes, unowned risks or mitigations not funded as incomplete. Show conflicts side by side without silent reconciliation and route them to Finance, accountable risk owners, Procurement and project owner.

ANALYSIS REQUIRED

Operation type. Calculation, estimate and comparison. **Decision interval.** At business-case, budget and material change checkpoints. **Minimum evidence.** Complete base cost, risk register with owners, evidence-based probability or scenario, consequence range, correlations, mitigations, treatment status and applicable Finance policy. **Method.** Separate base, known unknowns and excluded uncertainty; model correlated cost and schedule risk where evidence supports it; compare percentile allowance with policy and scenario stress. **Uncertainty.** Show P50 and selected confidence allowance, expected exposure, tail risk, double-count checks, correlation assumptions and risks excluded from quantification. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If risk ownership, base-cost boundary or correlation is unclear, provide a transparent scenario range and no precise allowance.

ANSWER BOUNDARY

Do not set contingency, recognise a provision, define risk appetite, certify adequacy or use an allowance to approve an unaffordable scope. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can see whether the allowance follows identified risks and where optimism, overlap or missing ownership distorts the case.

FP-074

STAKEHOLDER QUESTION

Are business-case benefits separated into cash-releasing, cost-avoidance, service, resilience, carbon and other outcomes with no overlap?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed energy, maintenance, availability, demand and service-condition telemetry may measure selected operational outcomes against a pre-agreed baseline. **Coverage, timing and aggregation.** Coverage must match the benefit owner, cost boundary, asset or service and time horizon; displaced demand, other budgets and wider outcomes remain explicit. Use aligned baseline and follow-up periods at the required cadence and retain changes in tariff, scope, weather and service. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Observed change does not establish cash release, causation, persistence, monetised value or benefit ownership.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns benefit taxonomy, cash treatment and ledger effect; project and service owners own benefit definitions and baselines; Energy or sustainability owns carbon; risk and continuity owners hold resilience outcomes; Procurement holds contract contribution. Pre-agreed definitions and authoritative financial evidence prevail over narrative or modelled benefit. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat benefits with no owner, baseline or counterfactual, savings already removed elsewhere, cost transferred between budgets, overlapping measures or outcome chosen after observation as unsupported. Show conflicts side by side without silent reconciliation and route them to Finance benefits owner and relevant project, service, Energy, sustainability or risk owners.

ANALYSIS REQUIRED

Operation type. Lookup, calculation and comparison. **Decision interval.** At business-case, delivery checkpoints and agreed benefit-realisation dates. **Minimum evidence.** Complete benefit register, category definitions, owner, baseline, target, timing, cost boundary, measurement method, counterfactual and links to any related benefit. **Method.** Create a benefit dependency map; reconcile cash claims to budgets, separate gross, net and displaced effects, and prevent one operational change appearing in multiple totals. **Uncertainty.** Show benefit by class, verified and forecast portions, overlap removed, confidence, timing and sensitivity to baseline and attribution. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a benefit lacks owner, boundary or evidence method, retain it as unquantified narrative and exclude it from totals.

ANSWER BOUNDARY

Do not certify benefits, remove budgets, monetise resilience or social outcomes without approved method, or treat carbon and cost as automatically additive. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Decision-makers can distinguish cash release, avoided future cost and non-financial outcomes without inflating the total case.

FP-075

STAKEHOLDER QUESTION

Which assumptions are decision-critical because plausible movement would make the case unaffordable, non-beneficial, undeliverable or incompatible with mandatory constraints?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed demand, asset performance, energy and service telemetry may calibrate selected operating assumptions within the measured boundary. **Coverage, timing and aggregation.** Coverage is limited to current assets, locations and regimes; market prices, approvals, legal constraints, supplier delivery and future service change remain external scenarios. Use representative intervals and preserve model inputs, baseline period and calibration error. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot validate mandatory compliance, future market behaviour or assumptions outside the observed range.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns affordability and financial assumptions; Procurement holds market and commercial inputs; project and service owners hold delivery and demand; competent, Legal and duty owners hold mandatory constraints; risk owners hold uncertainty ranges. Current authoritative constraints and owner assumptions prevail over model optimisation. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat point estimates without ranges, stale quotes, omitted dependencies, changed scope, unowned assumptions, benefits not evidenced or constraints represented as soft preferences as defects. Show conflicts side by side without silent reconciliation and route them to Business-case owner, Finance, Procurement, Legal and relevant competent owners.

ANALYSIS REQUIRED

Operation type. Calculation, comparison and diagnosis. **Decision interval.** At initial case, before commitment and after material change. **Minimum evidence.** Complete cash-flow and benefit model, mandatory constraints, owner-assigned ranges, dependency map, affordability limit, delivery schedule and decision criteria. **Method.** Run one-way and multi-way sensitivity, threshold and stress tests; identify combinations that cross affordability, value, schedule or mandatory gates and preserve non-linear interactions. **Uncertainty.** Show break-even values, probability or scenario of crossing each threshold, top sensitivity drivers, model coverage and assumptions not quantified. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If decision criteria or mandatory gates are undefined, report no pass or fail conclusion and request owner definitions.

ANSWER BOUNDARY

Do not redefine affordability, relax mandatory constraints, approve the case or use probability to override an accountable-owner decision. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The sponsor can focus evidence, negotiation and contingency on the assumptions capable of overturning the decision.

FP-076

STAKEHOLDER QUESTION

What are the cost, service and risk consequences of advancing, deferring, splitting or rephasing this purchase or project?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed asset condition, failure, demand, capacity and service-availability telemetry may inform timing consequences for mapped assets and services. **Coverage, timing and aggregation.** Coverage is limited to observable current conditions and demand; future failure, market lead time, interdependent projects and unmeasured service impact remain scenarios. Use monthly health and demand distributions with event evidence for material incidents. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Telemetry cannot authorise deferral, guarantee continued service or prove that acceleration will realise benefit earlier.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Project and service owners hold required date, dependencies and continuity; Finance owns funding profile and accounting impact; Procurement holds price validity, lead times and packaging; competent asset owners hold condition and safety; risk owners hold consequences. Mandatory dates, competent risk decisions and effective commercial terms prevail over modelled timing flexibility. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unknown dependencies, expired price, outage not confirmed, benefits without timing evidence, condition not current or split scope that changes competition or integration as unresolved. Show conflicts side by side without silent reconciliation and route them to Project owner, service owner, competent asset owner, Finance and Procurement.

ANALYSIS REQUIRED

Operation type. Forecast, calculation and comparison. **Decision interval.** At programme planning and whenever schedule or funding changes materially. **Minimum evidence.** Baseline schedule and cost, alternative dates or tranches, dependencies, funding limits, price and lead-time ranges, service and failure consequences, benefit timing and transition costs. **Method.** Model each timing scenario on a common horizon; calculate inflation, holding, mobilisation, outage, failure and delayed-benefit effects; include dependency and packaging changes. **Uncertainty.** Show net cost range, annual funding profile, service-risk exposure, completion range, lost or accelerated benefit and sensitivity to failure and market timing. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a mandatory date, competent risk limit or commercial validity is unknown, present scenario facts without a preferred timing.

ANSWER BOUNDARY

Do not rephase funding, commit a date, defer a safety action, reserve outages, vary scope or present a scenario as authority. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Leaders can compare the real cost and service consequences of timing choices rather than treating them as simple cash movements.

FP-077

STAKEHOLDER QUESTION

Which additional survey, meter, market test, pilot or clarification has positive decision value before the deadline, and which would not change the choice?

SENSORS AND TELEMETRY

Proposed modality and placement. Any proposed measurement must target a named decision uncertainty with a physically plausible modality, mapped location and pre-specified observation period. **Coverage, timing and aggregation.** Coverage must match the asset, service and scenario whose uncertainty affects the decision; generic extra sensing and person-level tracking are excluded. Use the minimum cadence and duration capable of testing the uncertainty, with a pre-agreed aggregation and stopping rule. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. For people-related proxies, keep aggregate non-identifying counts only. **Physical limit.** More data is not automatically better and cannot resolve legal, policy, owner or market questions outside its modality.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The decision owner and Finance hold choice criteria and materiality; Procurement owns permissible market engagement; competent owners specify surveys and pilots; data owners hold permissions; suppliers or surveyors provide evidence. Current authority, proportionality and competent survey design prevail over analytic curiosity. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat an evidence request with no decision link, no owner, no threshold, inaccessible location, insufficient time, market-contact risk or cost greater than plausible decision value as unjustified. Show conflicts side by side without silent reconciliation and route them to Decision owner, Finance, Procurement, competent survey or data owner.

ANALYSIS REQUIRED

Operation type. Estimate, comparison and recommendation limited to evidence acquisition. **Decision interval.** Before commitment and when a material assumption remains unresolved. **Minimum evidence.** Defined decision alternatives and thresholds, current uncertainty ranges, feasible evidence options, cost, duration, permission, expected accuracy and how results would change the choice. **Method.** Perform sensitivity first; where the choice can change, estimate expected value of information or use bounded decision-regret; compare with evidence cost and delay consequence. **Uncertainty.** Show maximum decision benefit, evidence cost, expected reduction in uncertainty, probability or scenario of changing choice and time penalty. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If the choice is insensitive or evidence cannot arrive in time, recommend no additional collection and state the residual uncertainty.

ANSWER BOUNDARY

Do not commission a survey, contact the market, deploy sensing, collect personal data or imply that more evidence transfers decision responsibility. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

The team can fund only evidence likely to change a material decision and avoid delay from low-value data collection.

FP-078

STAKEHOLDER QUESTION

What whole-life portfolio cost range follows when inflation, demand, failure, energy, supplier and delivery risks vary together?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed demand, failure, condition, energy and service telemetry may calibrate selected portfolio drivers for mapped assets and periods. **Coverage, timing and aggregation.** Coverage is limited to observed buildings, assets and failure modes; market, policy, supplier, construction and extreme-event risks require external scenarios. Use exposure-normalised monthly or seasonal distributions and retain cross-asset events and common operating conditions. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** Historical co-movement does not guarantee future dependence, capture unprecedented shocks or establish owner risk appetite.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns portfolio cost model, inflation and discount policy; asset and service owners hold condition, demand and consequence; Procurement holds supplier and market risks; Energy holds tariff and consumption; risk owners hold scenario dependencies. Current owner ranges and policy prevail over fitted distributions. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat inconsistent asset perimeter, double-counted risks, short history, structural changes, independent-risk assumptions without basis, stale prices or missing major projects as incomplete. Show conflicts side by side without silent reconciliation and route them to Finance portfolio owner, Procurement, Energy, asset and service risk owners.

ANALYSIS REQUIRED

Operation type. Forecast, estimate and comparison. **Decision interval.** Annually with refresh after material market, asset or programme change. **Minimum evidence.** Complete portfolio perimeter, base cash flows, owner ranges for principal drivers, dependency structure, common scenarios, timing and explicit exclusions. **Method.** Run correlated scenario or probabilistic simulation only where distributions are supportable; otherwise use coherent stress scenarios; reconcile portfolio outputs to asset-level inputs. **Uncertainty.** Show P10, P50 and P90 or low-base-high cost, tail exposure, correlation sensitivity, diversification effect, coverage and excluded risks. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If dependencies or portfolio perimeter are not supportable, provide scenario totals without probabilistic language and identify the missing structure.

ANSWER BOUNDARY

Do not set reserves, risk appetite or funding, certify probability, hide excluded assets or present a model tail as a guaranteed maximum. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Finance can see aggregate exposure when shared cost drivers deteriorate together rather than assuming independent best cases.

FP-079

STAKEHOLDER QUESTION

Which unknowns, stale sources, conflicting records and model limits must be shown beside the result before it is used?

SENSORS AND TELEMETRY

Proposed modality and placement. Every telemetry input used in the answer must expose modality, mapped asset or location, sampling, aggregation, calibration or validation, clock quality, uptime, gaps, drift and lineage. **Coverage, timing and aggregation.** Coverage is stated against the exact question perimeter and decision period; uninstrumented assets, blind spots and post-retrieval events remain explicit. Preserve source-native time and the aggregation used in the answer. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** No healthy sensor compensates for missing authority, stale commercial terms or a model applied outside its validated range.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Each domain owner remains authoritative for its source: Finance for financial records, Procurement for commercial records, Estates or IT for assets, Energy for meters, Legal and competent owners for interpretation and duties. Authoritative conflicts are displayed rather than resolved by the service. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unknown owner, permission not confirmed, effective date missing, observation outside period, retrieved data superseded, transformation without lineage or unresolved control total as a material limitation. Show conflicts side by side without silent reconciliation and route them to answer owner and every authoritative source owner.

ANALYSIS REQUIRED

Operation type. Diagnosis, lookup and recommendation limited to disclosure and next evidence action. **Decision interval.** With every answer and again immediately before a material decision. **Minimum evidence.** Source inventory, effective-observed-retrieved times, permissions, quality metrics, transformation lineage, validation range, conflict log, materiality threshold and named owner for each unresolved item. **Method.** Generate an evidence ledger; classify limitation by potential direction and maximum plausible effect; separate known unknown, unknown perimeter and model assumption; test whether the decision changes under each. **Uncertainty.** Show coverage, age, unresolved GBP or service scope, directional bias, model error and a decision-ready, conditional or not-ready status. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If a material source, owner or lineage is absent, suppress the affected conclusion and provide an evidence-gap statement.

ANSWER BOUNDARY

Do not conceal caveats, merge conflicting records, fabricate confidence, expose restricted source detail or shift accountability to the analytic service. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Decision-makers receive decision-changing limits at the same prominence as the result and can request the correct authoritative action.

FP-080

STAKEHOLDER QUESTION

Which forecast, cost, contract and evidence alerts are worth issuing, to which authorised roles, and under what suppression and expiry safeguards?

SENSORS AND TELEMETRY

Proposed modality and placement. Proposed meter, asset, work-order and service events may support alerts only for phenomena they directly cover; financial and commercial alerts primarily use authorised record changes. **Coverage, timing and aggregation.** Coverage must identify the cost line, contract, asset or location and permitted recipient group; blind spots and disconnected systems remain explicit. Use event or scheduled evaluation cadence matched to decision lead time, with deduplication and clock checks. **Verification, privacy and lineage.** Proposed only: use after deployment, mapping, commissioning, permission and live-health checks. Retain source/mapping ID, calibration/validation, clock offset, uptime, gaps, resets, drift and raw-to-aggregate lineage; exclude invalid intervals. No personal data is needed. **Physical limit.** An alert is not a diagnosis, approval, breach finding or instruction, and person-level behavioural alerts are excluded.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns cost and forecast thresholds; Procurement and contract owners hold notice dates and commercial triggers; asset, service, Energy and risk owners hold operational thresholds; data owners control recipients and permissions. Owner-approved thresholds and effective terms prevail over learned anomalies. **Time, permissions and provenance.** Retain owner, record/version ID, effective time, observed or transaction time, retrieval time, access basis and transformation lineage. **Staleness and conflict.** Treat unowned thresholds, stale recipient lists, duplicate sources, sensor health failure, contract trigger not effective, missing acknowledgement or model drift beyond validation as unsafe. Show conflicts side by side without silent reconciliation and route them to alert owner, authoritative domain owner, Finance or Procurement as applicable, and data-access owner.

ANALYSIS REQUIRED

Operation type. Observation, forecast, diagnosis and recommendation limited to alert routing. **Decision interval.** At the minimum cadence needed to preserve the action window, with periodic threshold confirmation. **Minimum evidence.** Named owner and recipient, decision lead time, materiality, authoritative source, threshold or model validation, expected base rate, escalation path, suppression rule and closure evidence. **Method.** Prefer deterministic triggers; validate predictive alerts on held-out history; group related events, suppress unhealthy inputs, require acknowledgement and measure precision, recall and time-to-action. **Uncertainty.** Show trigger basis, confidence, false-alarm rate, missed-event estimate, data health, materiality and expiry of the alert. Include coverage and unresolved exposure; avoid false precision. **Safe failure.** If owner, permission, threshold or source health fails, suppress the alert and record why rather than sending a speculative warning.

ANSWER BOUNDARY

Do not auto-approve, auto-pay, auto-order, auto-vary, contact a supplier, identify individuals, disclose restricted data or control building systems from an alert. Do not disclose restricted commercial or personal data; approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; certify value or compliance; authorise work; or control building systems. Finance, Procurement, Legal and the accountable owner retain authority.

WHY THIS IS USEFUL

Teams can receive fewer, earlier and more actionable warnings while preserving permissions, owner authority and alert credibility.

Eighty questions form eight connected decision areas, with ten records in each area.

FP-001 to FP-010

1. budgets, cost centres and expenditure context

Cost drivers, free balance, coding, classification, aggregation, recurring commitments, allocation, cut-off, eligibility and decision readiness.

FP-011 to FP-020

2. demand, specifications and purchase planning

Representative demand, alternatives, mandatory specifications, lead time, aggregation, interfaces, site evidence, stock, logistics and staged commitment.

FP-021 to FP-030

3. suppliers, tenders, contracts and service levels

Comparable service levels, supplier context, remedies, bid normalisation, volume terms, warranty, rates, delay dependencies, route choice and delivery model.

FP-031 to FP-040

4. work orders, invoices, variations and evidence

Order-to-invoice evidence, quantities, variations, duplicate candidates, stalled work, milestones, restoration, charging rules, queries and close-out.

FP-041 to FP-050

5. whole-life cost, maintenance and asset replacement

Asset priority, deterioration, remaining life, maintenance intervals, spares, whole-life options, deferred exposure, refurbishment, phasing and realised outcomes.

FP-051 to FP-060

6. utilities, energy, carbon and operating cost

Tariff and load drivers, weather normalisation, baseload, invoice reconciliation, control options, ventilation, water, investment, allocation and carbon lineage.

FP-061 to FP-070

7. compliance, risk, social value and responsible procurement

Authority evidence, supplier credentials, measurable commitments, conflicts, dependency, data duties, accessibility, end-of-life and attributable social value.

FP-071 to FP-080

8. forecasting, scenarios, benefits and uncertainty

Outturn, demand scenarios, contingency, benefit integrity, critical assumptions, timing, value of information, portfolio dependence, limits and alerts.

An answer advances only when evidence, permission, uncertainty and owner authority are explicit.

1 Deployment and mapping

Confirm the proposed modality exists, is mapped to the correct asset or location and covers the decision period.

2 Commissioning and health

Confirm commissioning, calibration or validation, clock alignment, uptime, gaps, resets and drift.

3 Permission and minimisation

Use the least granular permitted evidence; exclude identity, tracking and restricted content.

4 Authoritative sources

Name current owners and preserve precedence over telemetry, dashboards and historical spatial material.

5 Temporal alignment

Keep effective, observed or transaction, and retrieval times; prevent late or superseded evidence from appearing current.

6 Minimum evidence

State identity, coverage, comparison period, owner confirmation and any mandatory sample or completeness threshold.

7 Uncertainty and conflict

Quantify ranges, residuals, coverage and unresolved exposure; display conflicting records rather than merging them silently.

8 Safe failure

Suppress unsupported conclusions and route the next authoritative action when a material gate fails.

AUTHORITY REMAINS WITH ACCOUNTABLE PEOPLE

The service may organise permitted evidence and calculations. It does not approve or enact budgets, requisitions, purchase orders, tenders, supplier selection, invoices, payments, variations or contracts; disclose restricted commercial data; certify value or compliance; authorise work; or control systems. The May 2021 floor plan remains historical spatial reference only.